

Assignment Report

CS6903: Network Security

2025-26 - Semester II

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Task 1: Basic File Management Service

Summary and overview of TASK1

For Task 1, this is remote file management system using a Client-Server model. The aim here is to get two different processes talking to each other over a network. I used standard Python sockets to let a client send commands like LIST or INFO to a server, which then processes them one by one. Here the client acts as the interface for the user, and the server acts as the "manager" for the files stored in a specific directory.

Implementation Approach

To get the basic architecture running, I focused on setting up a stable connection before handling the specific commands:

Client Side:

Here I used `raw_input.split(maxsplit=1)` so the client can handle commands with or without arguments (like just LIST vs INFO file.txt). In this I have used the command part to `.upper()` to make sure the server recognizes it even if the user types in lowercase.

Server Side:

The server uses an if-elif block inside `process_command()` to route the request to the right helper function (like `handle_list` or `handle_info`).

Socket Backbone: I used the socket library to set up a TCP connection. On the server side, I used `setsockopt(SO_REUSEADDR)`. This is a life-saver because it lets me restart the server immediately without getting "Address already in use" errors. The server binds to port 5555 and listens for the client's `connect()` request.

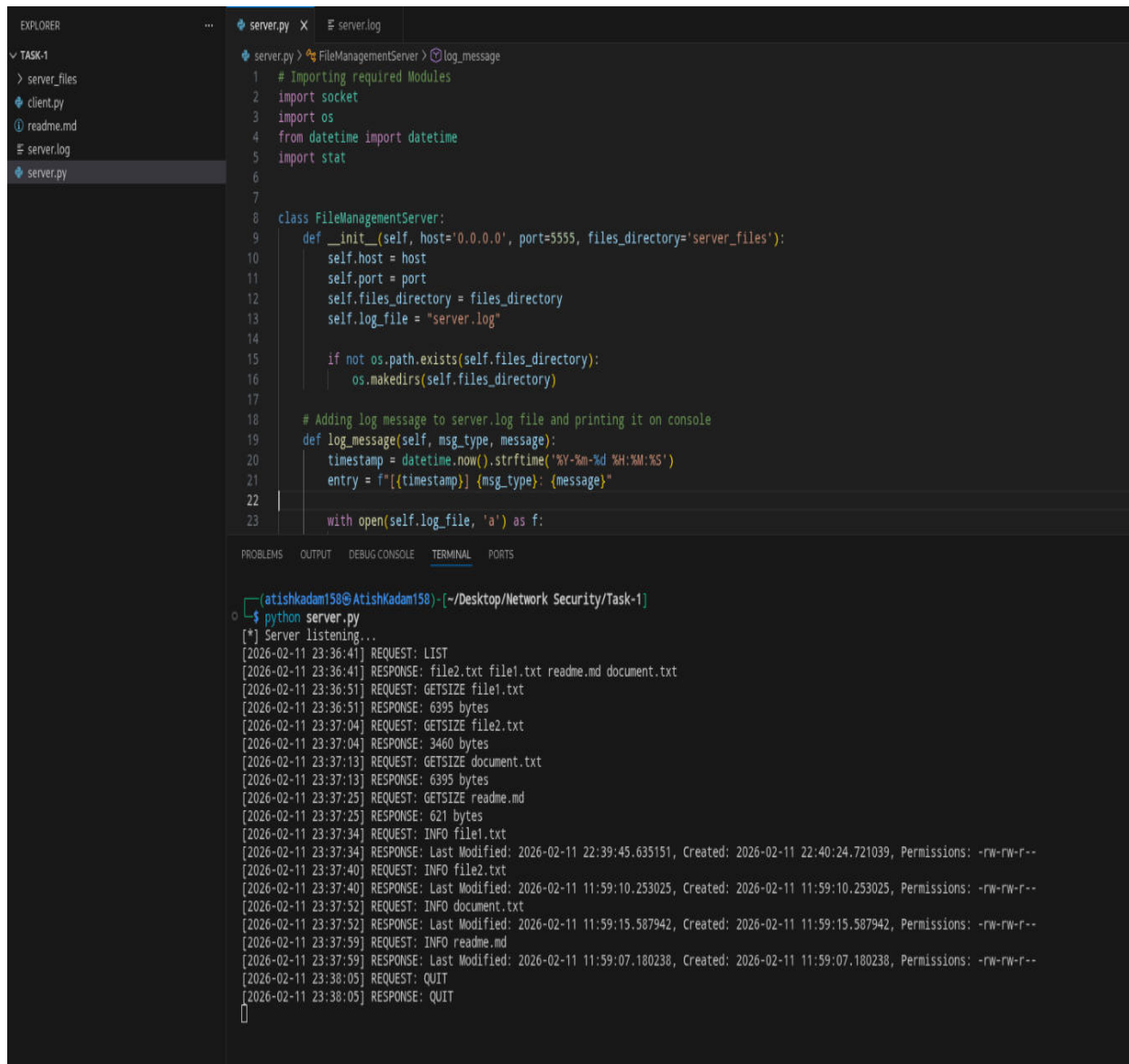
Logging System: I created a `log_message` function that writes every transaction to `server.log`. It captures the timestamp, whether it was a "REQUEST" or a "RESPONSE," and the actual data sent. It's set to append mode ('a'), so the history isn't wiped every time the server restarts.

Clean Exit: I implemented a QUIT command that sends a signal to both sides to break out of their loops and close the sockets properly, ensuring no "hanging" connections are left on the OS.

Server Side

This is the Task1 [server.py](#) file terminal capture screenshot.

In this screenshot, server Terminal shows REQUEST and RESPONSE of LIST, INFO, GETSIZE, and QUIT commands requested by user alongwith the timestamp.

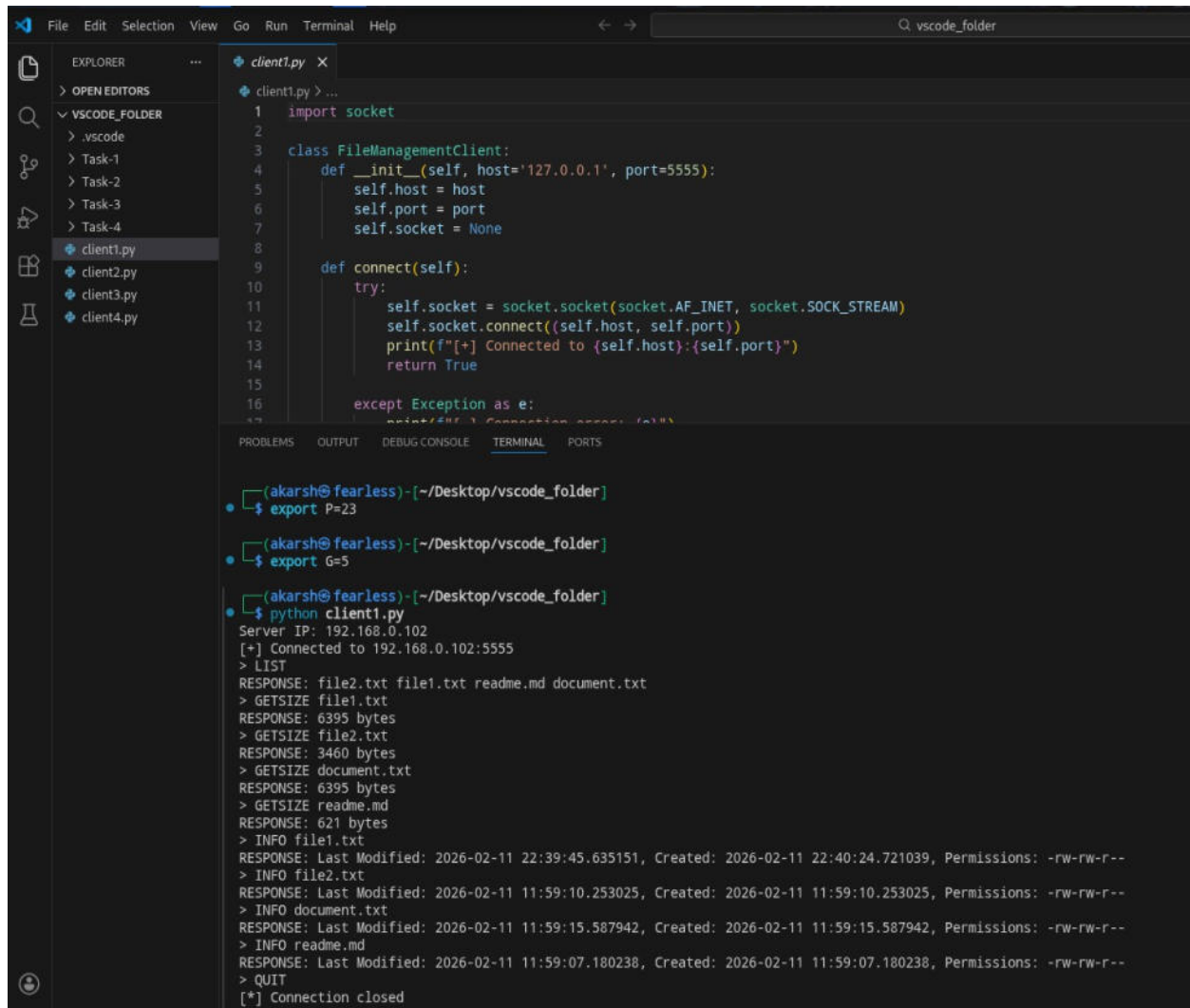


```
server.py X server.log
TASK-1
  > server_files
  > client.py
  > readme.md
  > server.log
  > server.py

server.py > FileManagementServer > log_message
1 # Importing required Modules
2 import socket
3 import os
4 from datetime import datetime
5 import stat
6
7
8 class FileManagementServer:
9     def __init__(self, host='0.0.0.0', port=5555, files_directory='server_files'):
10         self.host = host
11         self.port = port
12         self.files_directory = files_directory
13         self.log_file = "server.log"
14
15         if not os.path.exists(self.files_directory):
16             os.makedirs(self.files_directory)
17
18         # Adding log message to server.log file and printing it on console
19     def log_message(self, msg_type, message):
20         timestamp = datetime.now().strftime("%Y-%m-%d %H:%M:%S")
21         entry = f"[{timestamp}] {msg_type}: {message}"
22
23         with open(self.log_file, 'a') as f:
24
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
[atishkadam158@AtishKadam158] - [~/Desktop/Network Security/Task-1]
$ python server.py
[*] Server listening...
[2026-02-11 23:36:41] REQUEST: LIST
[2026-02-11 23:36:41] RESPONSE: file2.txt file1.txt readme.md document.txt
[2026-02-11 23:36:51] REQUEST: GETSIZE file1.txt
[2026-02-11 23:36:51] RESPONSE: 6395 bytes
[2026-02-11 23:37:04] REQUEST: GETSIZE file2.txt
[2026-02-11 23:37:04] RESPONSE: 3460 bytes
[2026-02-11 23:37:13] REQUEST: GETSIZE document.txt
[2026-02-11 23:37:13] RESPONSE: 6395 bytes
[2026-02-11 23:37:25] REQUEST: GETSIZE readme.md
[2026-02-11 23:37:25] RESPONSE: 621 bytes
[2026-02-11 23:37:34] REQUEST: INFO file1.txt
[2026-02-11 23:37:34] RESPONSE: Last Modified: 2026-02-11 22:39:45.635151, Created: 2026-02-11 22:40:24.721039, Permissions: -rw-rw-r--
[2026-02-11 23:37:40] REQUEST: INFO file2.txt
[2026-02-11 23:37:40] RESPONSE: Last Modified: 2026-02-11 11:59:10.253025, Created: 2026-02-11 11:59:10.253025, Permissions: -rw-rw-r--
[2026-02-11 23:37:52] REQUEST: INFO document.txt
[2026-02-11 23:37:52] RESPONSE: Last Modified: 2026-02-11 11:59:15.587942, Created: 2026-02-11 11:59:15.587942, Permissions: -rw-rw-r--
[2026-02-11 23:37:59] REQUEST: INFO readme.md
[2026-02-11 23:37:59] RESPONSE: Last Modified: 2026-02-11 11:59:07.180238, Created: 2026-02-11 11:59:07.180238, Permissions: -rw-rw-r--
[2026-02-11 23:38:05] REQUEST: QUIT
[2026-02-11 23:38:05] RESPONSE: QUIT
[]
```

Client side

This is the TASK1 [client.py](#) file terminal capture screenshot. In this screenshot , the client got the response of input command given to server in this format as like of LIST command it shows as RESPONSE: file2.txt file1.txt readme.md document.txt .Similarly the client got the response of other commands of GETSIZE,INFO and QUIT .



The screenshot shows a VS Code editor with the file `client.py` open. The file contains a Python class `FileManagementClient` with methods `__init__`, `connect`, and `connect`. The terminal output shows the client's interaction with the server.

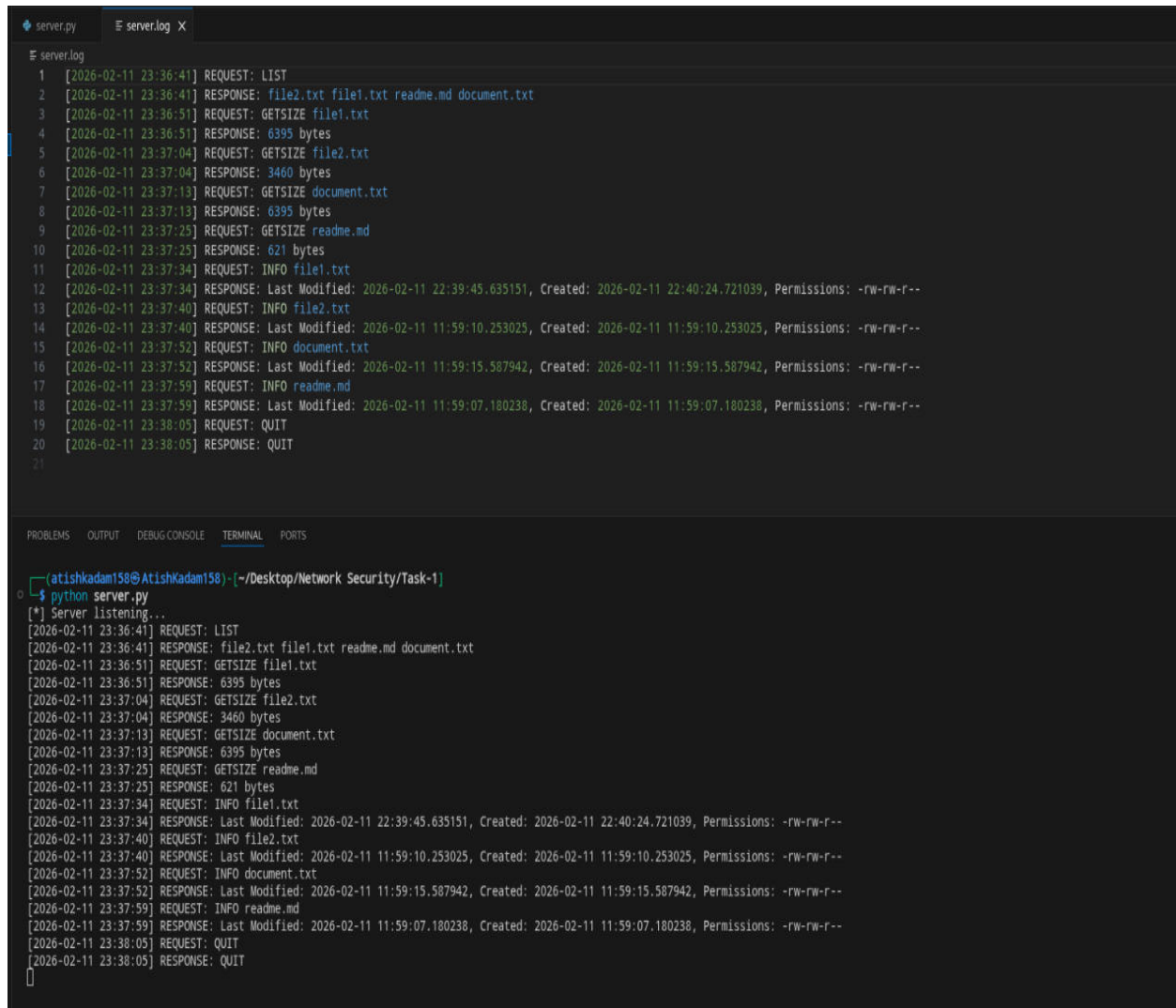
```
1 import socket
2
3 class FileManagementClient:
4     def __init__(self, host='127.0.0.1', port=5555):
5         self.host = host
6         self.port = port
7         self.socket = None
8
9     def connect(self):
10        try:
11            self.socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
12            self.socket.connect((self.host, self.port))
13            print(f"[+] Connected to {self.host}:{self.port}")
14            return True
15        except Exception as e:
16            print(f"[-] Connection error: {e}")
```

Terminal Output:

```
(akarsh@fearless)-[~/Desktop/vscode_folder]
$ export P=23
(akarsh@fearless)-[~/Desktop/vscode_folder]
$ export G=5
(akarsh@fearless)-[~/Desktop/vscode_folder]
$ python client1.py
Server IP: 192.168.0.102
[+] Connected to 192.168.0.102:5555
> LIST
RESPONSE: file2.txt file1.txt readme.md document.txt
> GETSIZE file1.txt
RESPONSE: 6395 bytes
> GETSIZE file2.txt
RESPONSE: 3460 bytes
> GETSIZE document.txt
RESPONSE: 6395 bytes
> GETSIZE readme.md
RESPONSE: 621 bytes
> INFO file1.txt
RESPONSE: Last Modified: 2026-02-11 22:39:45.635151, Created: 2026-02-11 22:40:24.721039, Permissions: -rw-rw-r--
> INFO file2.txt
RESPONSE: Last Modified: 2026-02-11 11:59:10.253025, Created: 2026-02-11 11:59:10.253025, Permissions: -rw-rw-r--
> INFO document.txt
RESPONSE: Last Modified: 2026-02-11 11:59:15.587942, Created: 2026-02-11 11:59:15.587942, Permissions: -rw-rw-r--
> INFO readme.md
RESPONSE: Last Modified: 2026-02-11 11:59:07.180238, Created: 2026-02-11 11:59:07.180238, Permissions: -rw-rw-r--
> QUIT
[*] Connection closed
```

Server.log

This is the TASK 1 **server.log** file ,which shows about the activity performed at server by client. It contains the timestamp at which the client asked for the command and at what time the response was received at client.



The image shows a code editor with two tabs: 'server.py' and 'server.log'. The 'server.log' tab is active, displaying a list of 20 log entries. Each entry consists of a line number, a timestamp in brackets, and a log message. The messages are categorized as REQUEST or RESPONSE, detailing file operations like LIST, GETSIZE, INFO, and QUIT, along with file names and metadata like Last Modified, Created, and Permissions.

```
1 [2026-02-11 23:36:41] REQUEST: LIST
2 [2026-02-11 23:36:41] RESPONSE: file2.txt file1.txt readme.md document.txt
3 [2026-02-11 23:36:51] REQUEST: GETSIZE file1.txt
4 [2026-02-11 23:36:51] RESPONSE: 6395 bytes
5 [2026-02-11 23:37:04] REQUEST: GETSIZE file2.txt
6 [2026-02-11 23:37:04] RESPONSE: 3460 bytes
7 [2026-02-11 23:37:13] REQUEST: GETSIZE document.txt
8 [2026-02-11 23:37:13] RESPONSE: 6395 bytes
9 [2026-02-11 23:37:25] REQUEST: GETSIZE readme.md
10 [2026-02-11 23:37:25] RESPONSE: 621 bytes
11 [2026-02-11 23:37:34] REQUEST: INFO file1.txt
12 [2026-02-11 23:37:34] RESPONSE: Last Modified: 2026-02-11 22:39:45.635151, Created: 2026-02-11 22:40:24.721039, Permissions: -rw-rw-r--
13 [2026-02-11 23:37:40] REQUEST: INFO file2.txt
14 [2026-02-11 23:37:40] RESPONSE: Last Modified: 2026-02-11 11:59:10.253025, Created: 2026-02-11 11:59:10.253025, Permissions: -rw-rw-r--
15 [2026-02-11 23:37:52] REQUEST: INFO document.txt
16 [2026-02-11 23:37:52] RESPONSE: Last Modified: 2026-02-11 11:59:15.587942, Created: 2026-02-11 11:59:15.587942, Permissions: -rw-rw-r--
17 [2026-02-11 23:37:59] REQUEST: INFO readme.md
18 [2026-02-11 23:37:59] RESPONSE: Last Modified: 2026-02-11 11:59:07.180238, Created: 2026-02-11 11:59:07.180238, Permissions: -rw-rw-r--
19 [2026-02-11 23:38:05] REQUEST: QUIT
20 [2026-02-11 23:38:05] RESPONSE: QUIT
21
```

Below the code editor, a terminal window is open, showing the command prompt and the execution of 'python server.py'. The terminal output mirrors the log file content, starting with 'Server listening...' and then displaying the same sequence of requests and responses.

```
(atishkadam158@AtishKadam158) ~/Desktop/Network Security/Task-1
$ python server.py
[*] Server listening...
[2026-02-11 23:36:41] REQUEST: LIST
[2026-02-11 23:36:41] RESPONSE: file2.txt file1.txt readme.md document.txt
[2026-02-11 23:36:51] REQUEST: GETSIZE file1.txt
[2026-02-11 23:36:51] RESPONSE: 6395 bytes
[2026-02-11 23:37:04] REQUEST: GETSIZE file2.txt
[2026-02-11 23:37:04] RESPONSE: 3460 bytes
[2026-02-11 23:37:13] REQUEST: GETSIZE document.txt
[2026-02-11 23:37:13] RESPONSE: 6395 bytes
[2026-02-11 23:37:25] REQUEST: GETSIZE readme.md
[2026-02-11 23:37:25] RESPONSE: 621 bytes
[2026-02-11 23:37:34] REQUEST: INFO file1.txt
[2026-02-11 23:37:34] RESPONSE: Last Modified: 2026-02-11 22:39:45.635151, Created: 2026-02-11 22:40:24.721039, Permissions: -rw-rw-r--
[2026-02-11 23:37:40] REQUEST: INFO file2.txt
[2026-02-11 23:37:40] RESPONSE: Last Modified: 2026-02-11 11:59:10.253025, Created: 2026-02-11 11:59:10.253025, Permissions: -rw-rw-r--
[2026-02-11 23:37:52] REQUEST: INFO document.txt
[2026-02-11 23:37:52] RESPONSE: Last Modified: 2026-02-11 11:59:15.587942, Created: 2026-02-11 11:59:15.587942, Permissions: -rw-rw-r--
[2026-02-11 23:37:59] REQUEST: INFO readme.md
[2026-02-11 23:37:59] RESPONSE: Last Modified: 2026-02-11 11:59:07.180238, Created: 2026-02-11 11:59:07.180238, Permissions: -rw-rw-r--
[2026-02-11 23:38:05] REQUEST: QUIT
[2026-02-11 23:38:05] RESPONSE: QUIT
```

Task 2: Authentication with Multiple Clients and Session Key Establishment

Summary and Design Overview

The objective of this task is to transition from a basic service to a secure, multi-user environment. The design focuses on three core security pillars:

Concurrency: Utilizing threading to handle multiple clients simultaneously.

Authentication: Implementing a Challenge-Response protocol using nonces and SHA-256 to prevent password sniffing.

Key Exchange: Establishing a symmetric session key via the Diffie-Hellman protocol to secure future communications.

Implementation Approach

Based on the submitted code, the implementation follows these steps:

- **Multi-Client Support:** The server uses the threading library. For every successful `accept()` call, a new thread is spawned to handle that specific client's session, ensuring the server remains responsive to other incoming connections.
- **Authentication Protocol:**
 - The client sends a username.
 - The server checks `credentials.txt`. If the user exists, it generates a cryptographically secure random **nonce** and sends it to the client. The client concatenates the nonce and their `shared_secret`, then sends the SHA-256 hash back: `HASH(nonce || shared_secret)`
 - The server performs the same calculation; if they match, access is granted.

Diffie-Hellman (DH) Key Exchange:

- The client pulls `P` and `G` from environment variables.
- Both parties generate private keys (`a` and `b`) and exchange public values ($A = G^a \mod P$ and $B = G^b \mod P$).
- The final shared secret is derived as $K = B^a \mod P$ (client) and $K = A^b \mod P$ (server).
- **Account Lockout Policy:** If a username fails three consecutive times, the server logs the event, blocks the username for that session, and closes the socket.

Server.log

This is the **Task2 server.log** file terminal capture screenshot.

In this basically after implementation of TASK1 we have made this task as TASK2 which consist of added authentication ,multi client support as well as secure key establishment.

Here after the matching of username and shared secret ,this shows “**AUTH SUCCESS**” which means that the client1 name “alice” and client2 name “charlie” is now authenticated to server .

Later in order to establish the session ,

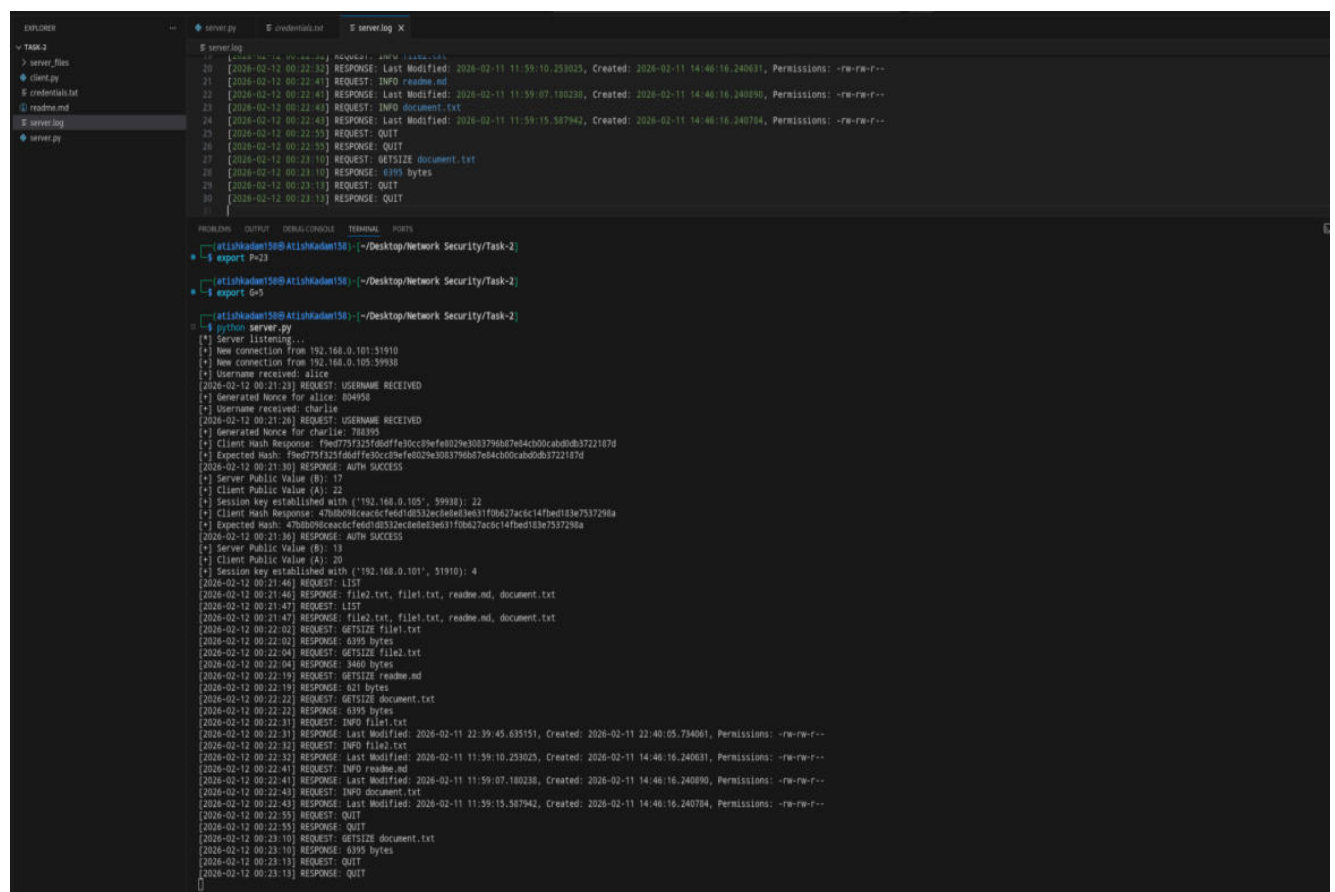
the client A public value and server B public value of both the client is computed to make the secret key for the session establishment and

show the output as “**Session key established with ('192.168.0.105',59938)** for client2

and “**Session key established with ('192.168.0.101', 51910)** for client 1

Further this task gives the response asked by client1 and 2 for the commands like LIST, GETSIZE file1.txt, INFO file1.txt, QUIT command same as done in TASK1 but here we have done the same for multiclient (192.168.0.105 , 59938 as client 2)
(192.168.0.101, 51910 as client 1)

Server side



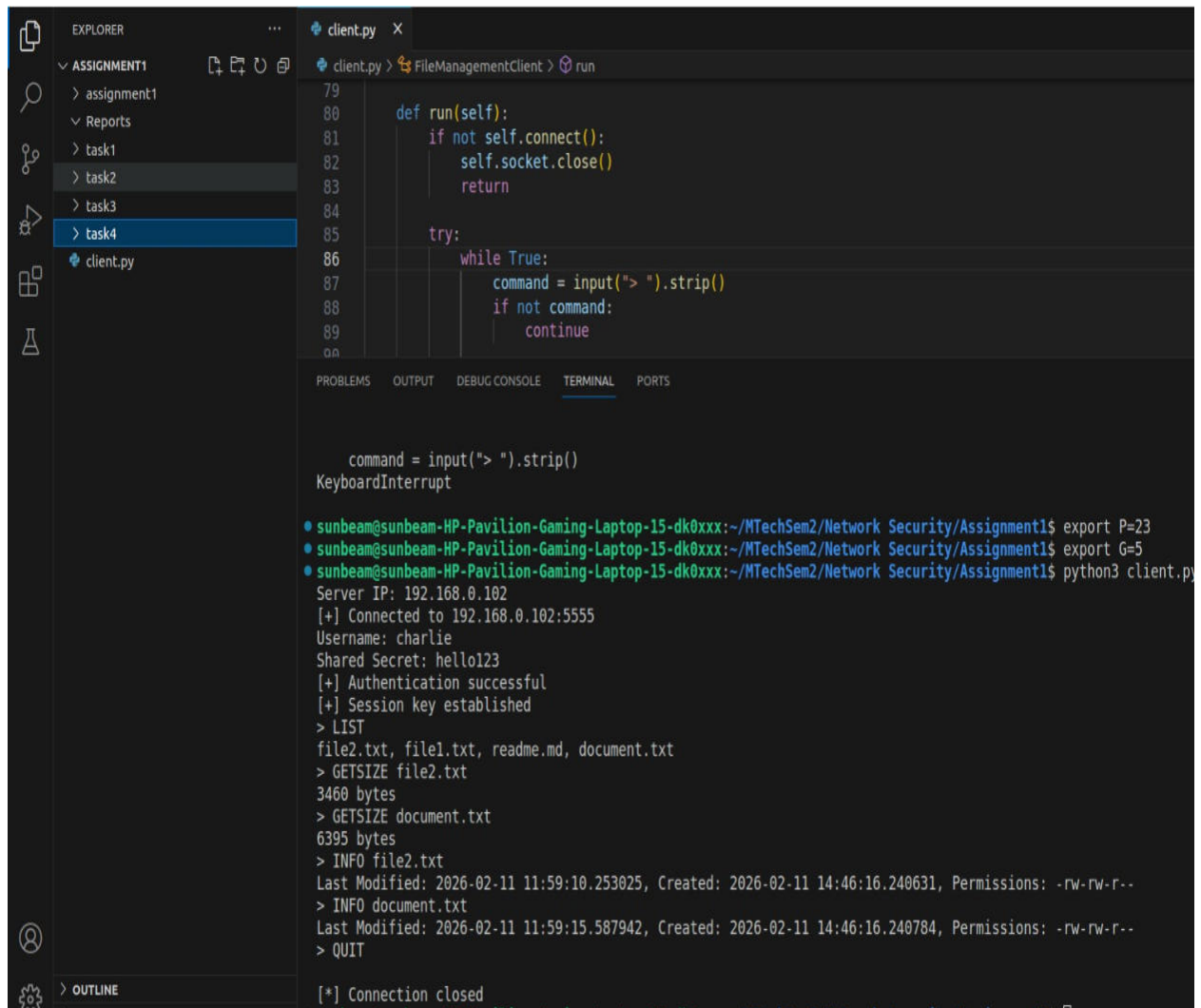
```
server.py
20  [{"sha256": "0x1234567890abcdef", "name": "file1.txt"}]
21  [2026-02-12 00:22:32] RESPONSE: Last Modified: 2026-02-11 11:59:10.253025, Created: 2026-02-11 14:40:16.240631, Permissions: -rw-rw-r--
22  [2026-02-12 00:22:41] REQUEST: INFO readme.md
23  [2026-02-12 00:22:41] RESPONSE: Last Modified: 2026-02-11 11:59:07.180238, Created: 2026-02-11 14:40:16.240890, Permissions: -rw-rw-r--
24  [2026-02-12 00:22:43] REQUEST: INFO document.txt
25  [2026-02-12 00:22:43] RESPONSE: Last Modified: 2026-02-11 11:59:15.587942, Created: 2026-02-11 14:40:16.240704, Permissions: -rw-rw-r--
26  [2026-02-12 00:22:55] REQUEST: QUIT
27  [2026-02-12 00:22:55] RESPONSE: QUIT
28  [2026-02-12 00:23:10] REQUEST: GETSIZE document.txt
29  [2026-02-12 00:23:10] RESPONSE: 6395 bytes
30  [2026-02-12 00:23:13] REQUEST: QUIT
31  [2026-02-12 00:23:13] RESPONSE: QUIT
32  []

PROBLEMS OUTPUT DEBUG/CONSOLE TERMINAL PORTS
[+] atishkadam150@atishkadam150: ~/Desktop/Network Security/Task-2
$ export P=23
[+] atishkadam150@atishkadam150: ~/Desktop/Network Security/Task-2
$ export G=5
[+] atishkadam150@atishkadam150: ~/Desktop/Network Security/Task-2
$ python server.py
[*] Server listening...
[*] New connection from 192.168.0.101:51910
[*] New connection from 192.168.0.105:59938
[*] Username received: alice
[2026-02-12 00:21:23] REQUEST: USERNAME RECEIVED
[*] Generated nonce for alice: 804950
[*] Username received: charlie
[2026-02-12 00:21:26] REQUEST: USERNAME RECEIVED
[*] Generated nonce for charlie: 788395
[*] Client Hash Response: fbed775f25f0d0ffef0cc9bfe029e3083796a87e04c30cab0db3722187d
[*] Expected Hash: fbed775f25f0d0ffef0cc9bfe029e3083796a87e04c30cab0db3722187d
[2026-02-12 00:21:30] RESPONSE: AUTH SUCCESS
[*] Server Public Value (B): 17
[*] Client Public Value (A): 22
[*] Session key established with ('192.168.0.105', 59938): 22
[*] Client Hash Response: 47db090ceac3cfedd0532c2ed0e31f0b627ac9c14fbed183e7537298a
[*] Expected Hash: 47db090ceac3cfedd0532c2ed0e31f0b627ac9c14fbed183e7537298a
[2026-02-12 00:21:36] RESPONSE: AUTH SUCCESS
[*] Server Public Value (B): 13
[*] Client Public Value (A): 20
[*] Session key established with ('192.168.0.101', 51910): 4
[2026-02-12 00:21:46] REQUEST: LIST
[2026-02-12 00:21:46] RESPONSE: file2.txt, file1.txt, readme.md, document.txt
[2026-02-12 00:21:47] REQUEST: LIST
[2026-02-12 00:21:47] RESPONSE: file2.txt, file1.txt, readme.md, document.txt
[2026-02-12 00:22:02] REQUEST: GETSIZE file1.txt
[2026-02-12 00:22:02] RESPONSE: 6395 bytes
[2026-02-12 00:22:04] REQUEST: GETSIZE file2.txt
[2026-02-12 00:22:04] RESPONSE: 3486 bytes
[2026-02-12 00:22:19] REQUEST: GETSIZE readme.md
[2026-02-12 00:22:19] RESPONSE: 621 bytes
[2026-02-12 00:22:21] REQUEST: GETSIZE document.txt
[2026-02-12 00:22:22] RESPONSE: 6395 bytes
[2026-02-12 00:22:31] REQUEST: INFO file1.txt
[2026-02-12 00:22:31] RESPONSE: Last Modified: 2026-02-11 22:39:45.635151, Created: 2026-02-11 22:40:05.734061, Permissions: -rw-rw-r--
[2026-02-12 00:22:32] REQUEST: INFO file2.txt
[2026-02-12 00:22:32] RESPONSE: Last Modified: 2026-02-11 11:59:10.253025, Created: 2026-02-11 14:40:16.240631, Permissions: -rw-rw-r--
[2026-02-12 00:22:41] REQUEST: INFO readme.md
[2026-02-12 00:22:41] RESPONSE: Last Modified: 2026-02-11 11:59:07.180238, Created: 2026-02-11 14:40:16.240890, Permissions: -rw-rw-r--
[2026-02-12 00:22:43] REQUEST: INFO document.txt
[2026-02-12 00:22:43] RESPONSE: Last Modified: 2026-02-11 11:59:15.587942, Created: 2026-02-11 14:40:16.240704, Permissions: -rw-rw-r--
[2026-02-12 00:22:55] REQUEST: QUIT
[2026-02-12 00:22:55] RESPONSE: QUIT
[2026-02-12 00:23:10] REQUEST: GETSIZE document.txt
[2026-02-12 00:23:10] RESPONSE: 6395 bytes
[2026-02-12 00:23:13] REQUEST: QUIT
[2026-02-12 00:23:13] RESPONSE: QUIT
[]
```

Client 1 side (as multiuser)

This is the capture of Client 1 (as multiclient user) for TASK2 with IP 192.168.0.101 connected with server IP=192.168.0.102'

Here the client 1 (multiuser) is “charlie” with it “shared secret key : hello123” has been authenticated by server as we can see in terminal it shows “Authentication Successful” and further the session key established and after which only the client 1 performs the command as LIST,GETSIZE file2.txt etc as it can be seen in the terminal capture below.



The screenshot displays a VS Code interface with a file explorer on the left showing a project structure with folders 'ASSIGNMENT1', 'Reports', and 'tasks' (task1, task2, task3, task4), and a file 'client.py'. The main editor window shows the code for 'client.py', which includes a 'run' method for a 'FileManagementClient' class. The terminal window at the bottom shows the execution of the client, including connection details, authentication success, and file management commands like 'LIST', 'GETSIZE', and 'QUIT'.

```
client.py
79
80     def run(self):
81         if not self.connect():
82             self.socket.close()
83             return
84
85         try:
86             while True:
87                 command = input("> ").strip()
88                 if not command:
89                     continue
90
91
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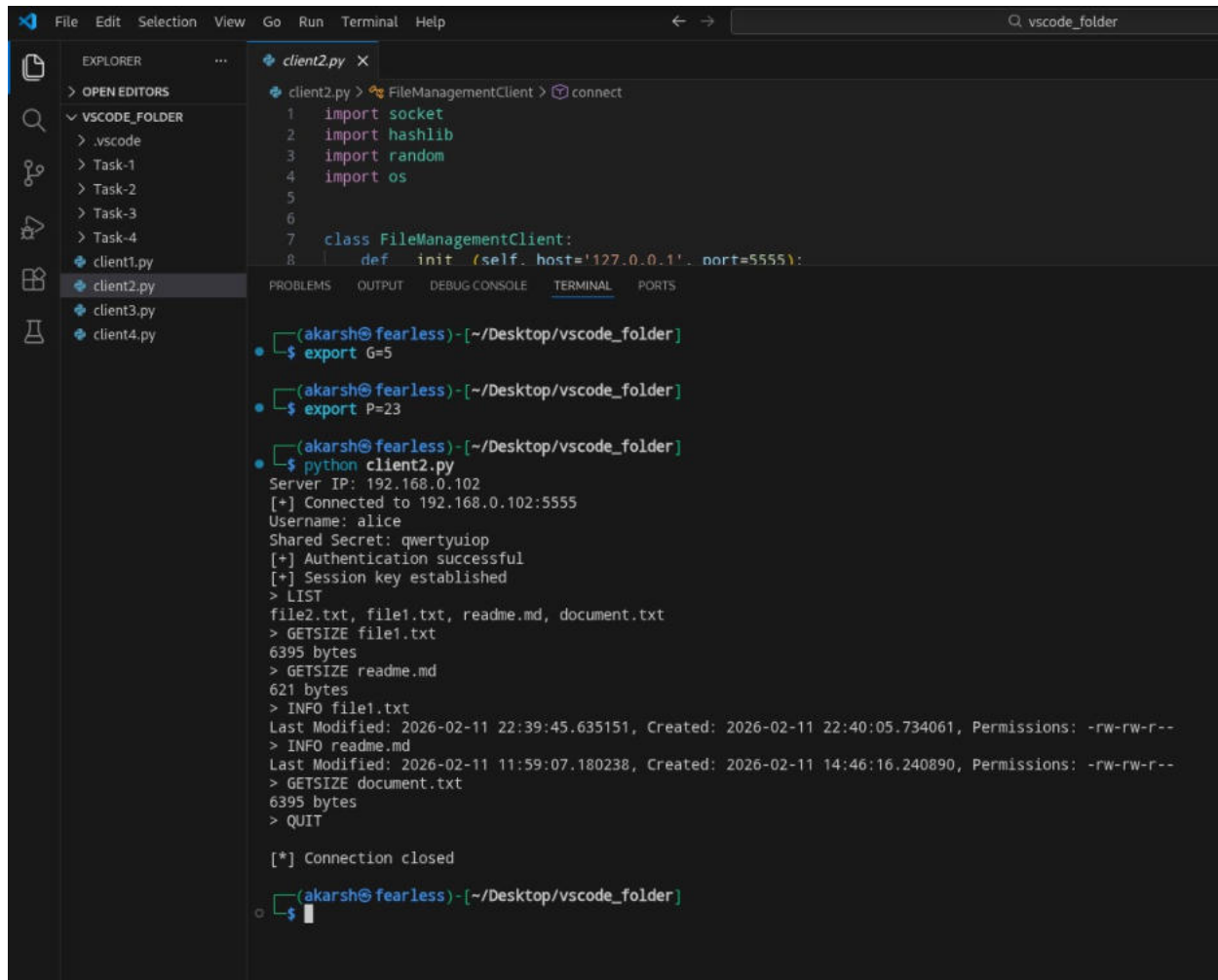
```
command = input("> ").strip()
KeyboardInterrupt

sunbeam@sunbeam-HP-Pavilion-Gaming-Laptop-15-dk0xxx:~/MTechSem2/Network Security/Assignment1$ export P=23
sunbeam@sunbeam-HP-Pavilion-Gaming-Laptop-15-dk0xxx:~/MTechSem2/Network Security/Assignment1$ export G=5
sunbeam@sunbeam-HP-Pavilion-Gaming-Laptop-15-dk0xxx:~/MTechSem2/Network Security/Assignment1$ python3 client.py
Server IP: 192.168.0.102
[+] Connected to 192.168.0.102:5555
Username: charlie
Shared Secret: hello123
[+] Authentication successful
[+] Session key established
> LIST
file2.txt, file1.txt, readme.md, document.txt
> GETSIZE file2.txt
3460 bytes
> GETSIZE document.txt
6395 bytes
> INFO file2.txt
Last Modified: 2026-02-11 11:59:10.253025, Created: 2026-02-11 14:46:16.240631, Permissions: -rw-rw-r--
> INFO document.txt
Last Modified: 2026-02-11 11:59:15.587942, Created: 2026-02-11 14:46:16.240784, Permissions: -rw-rw-r--
> QUIT
[*] Connection closed
```


Client 2 side (as multiuser)

This is the capture of Client 2 (as multiclient user) for TASK2 with IP 192.168.0.105 connected with server IP=192.168.0.102

Here the client2 (multiuser) is “alice” with it “shared secret key : qwertyuiop” has been authenticated by server as we can see in terminal it shows “Authentication Successful” and further the session key established with server and after which only the client2 performs the command as LIST,GETSIZE file1.txt etc as it can be seen in the terminal capture below.



```
File Edit Selection View Go Run Terminal Help
client2.py X
client2.py > FileManagementClient > connect
1 import socket
2 import hashlib
3 import random
4 import os
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7 class FileManagementClient:
8     def init (self, host='127.0.0.1', port=5555):
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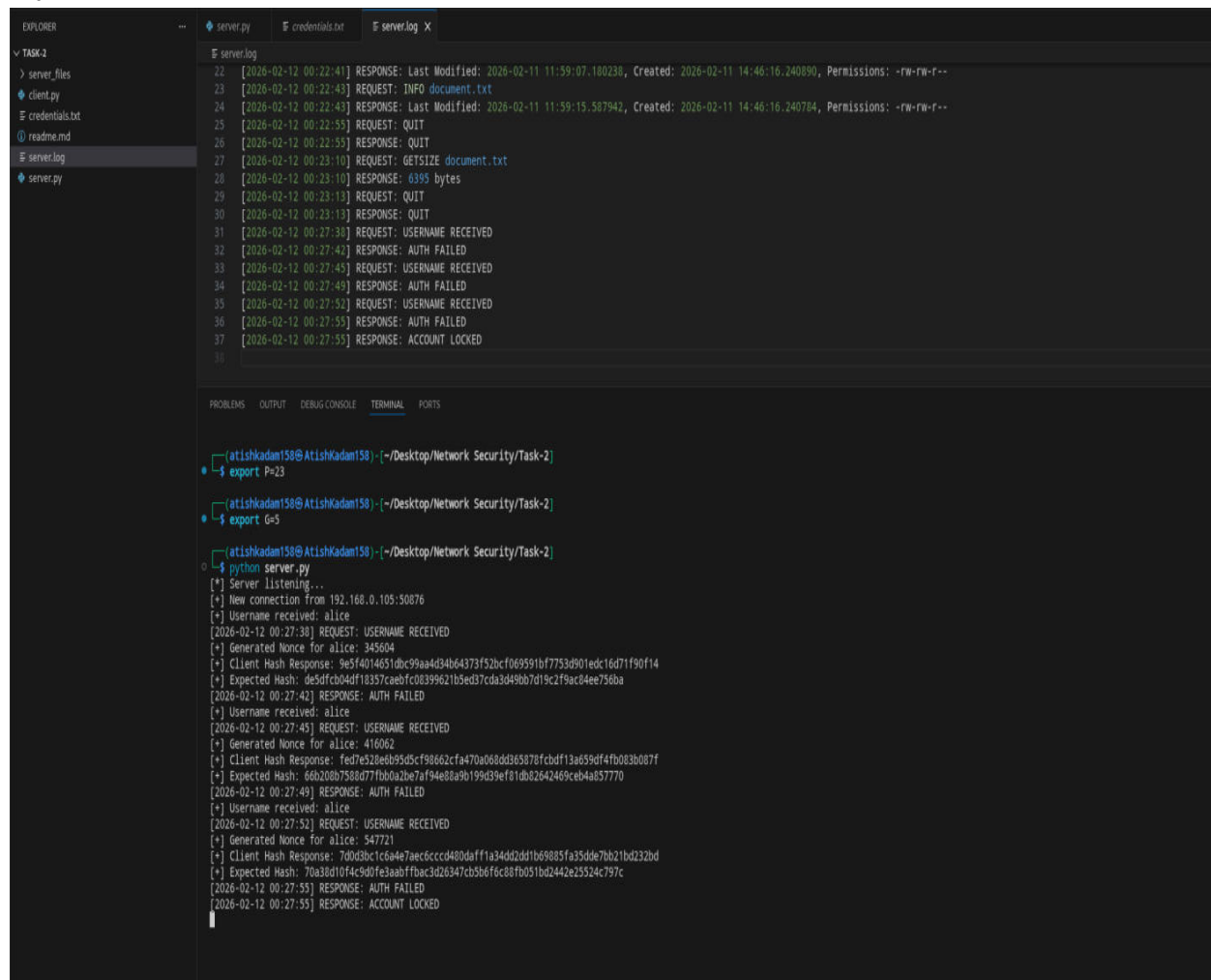

Server side(terminal during lockout)

Now here in **TASK2** of **server.log** file terminal capture

As per **Account Lockout Policy** we have demonstrated how the account gets blocked by the client when the credential does not match as of server credential.

Here is the capture of the server terminal when the account gets blocked .

Here the client with IP=192.168.0.105 is blocked by server for entering the wrong shared secret key.



```
server.log
22 [2026-02-12 00:22:41] RESPONSE: Last Modified: 2026-02-11 11:59:07.180236, Created: 2026-02-11 14:46:16.240890, Permissions: -rw-rw-r--
23 [2026-02-12 00:22:43] REQUEST: INFO document.txt
24 [2026-02-12 00:22:43] RESPONSE: Last Modified: 2026-02-11 11:59:15.587942, Created: 2026-02-11 14:46:16.240784, Permissions: -rw-rw-r--
25 [2026-02-12 00:22:55] REQUEST: QUIT
26 [2026-02-12 00:22:55] RESPONSE: QUIT
27 [2026-02-12 00:23:10] REQUEST: GETSIZE document.txt
28 [2026-02-12 00:23:10] RESPONSE: 6395 bytes
29 [2026-02-12 00:23:13] REQUEST: QUIT
30 [2026-02-12 00:23:13] RESPONSE: QUIT
31 [2026-02-12 00:27:38] REQUEST: USERNAME RECEIVED
32 [2026-02-12 00:27:42] RESPONSE: AUTH FAILED
33 [2026-02-12 00:27:45] REQUEST: USERNAME RECEIVED
34 [2026-02-12 00:27:49] RESPONSE: AUTH FAILED
35 [2026-02-12 00:27:52] REQUEST: USERNAME RECEIVED
36 [2026-02-12 00:27:55] RESPONSE: AUTH FAILED
37 [2026-02-12 00:27:55] RESPONSE: ACCOUNT LOCKED
38

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

[atishkadam158@AtishKadam158] ~/Desktop/Network Security/Task-2
+ export P=23

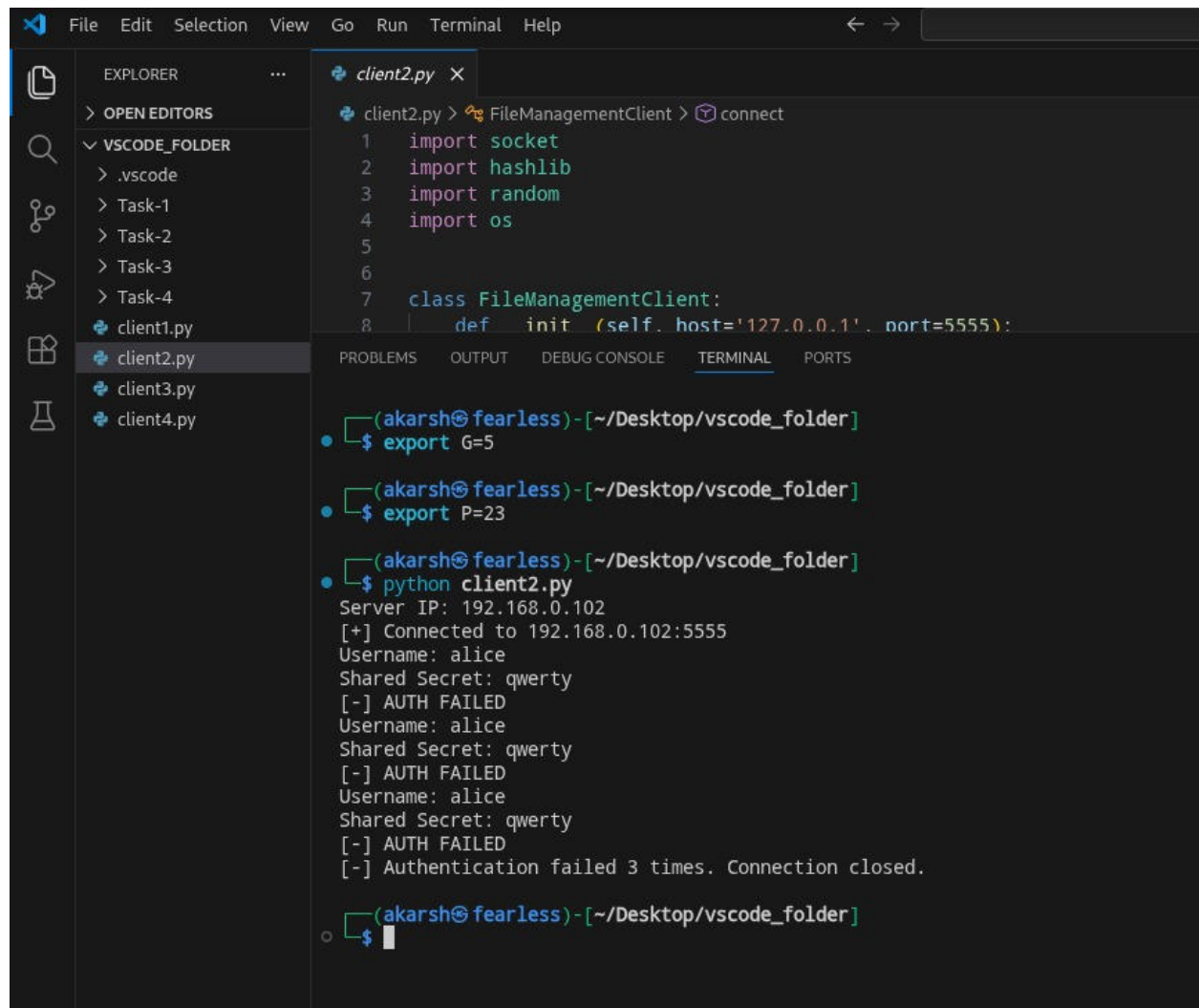
[atishkadam158@AtishKadam158] ~/Desktop/Network Security/Task-2
+ export G=5

[atishkadam158@AtishKadam158] ~/Desktop/Network Security/Task-2
+ python server.py
[*] Server listening...
[*] New connection from 192.168.0.105:50876
[*] Username received: alice
[2026-02-12 00:27:38] REQUEST: USERNAME RECEIVED
[*] Generated Nonce for alice: 345604
[*] Client Hash Response: 9e5f4014651dbc99aa4d34b64373f52bcf069591bf7753d901edc16d71f90f14
[*] Expected Hash: de5dfcb04df18357caebfc08399621b5ed37cda3d49bb7d19c2f9ac04ee756ba
[2026-02-12 00:27:42] RESPONSE: AUTH FAILED
[*] Username received: alice
[2026-02-12 00:27:45] REQUEST: USERNAME RECEIVED
[*] Generated Nonce for alice: 416062
[*] Client Hash Response: fed7e528eb95d5cf98662cfa470a068dd363878fcbdf13a659df4fb083b087f
[*] Expected Hash: 66b208b7588d77fbb0a2be7af94e88a9b199d39ef81db82642469ceb4a857770
[2026-02-12 00:27:49] RESPONSE: AUTH FAILED
[*] Username received: alice
[2026-02-12 00:27:52] REQUEST: USERNAME RECEIVED
[*] Generated Nonce for alice: 547721
[*] Client Hash Response: 70d3b01c6a4e7a6c6cccd480daff1a34dd2dd1b69885fa35dde7bb21bd232bd
[*] Expected Hash: 70a38d10f4c9d0fe3aabffbac3d26347cb3b6f6c88fb051bd2442e25524c797c
[2026-02-12 00:27:55] RESPONSE: AUTH FAILED
[2026-02-12 00:27:55] RESPONSE: ACCOUNT LOCKED
```

Client side (terminal during logout)

This is the [client2.py](#) file terminal capture

In this it shows how the client with IP =192.168.0.105 name “alice” enters the username on server ip= 192.168.0.102 but enters the wrong shared secret key consecutive 3 times which makes the server to block the account.



```
client2.py > FileManagementClient > connect
1  import socket
2  import hashlib
3  import random
4  import os
5
6
7  class FileManagementClient:
8      def __init__(self, host='127.0.0.1', port=5555):

(akarsh@fearless)-[~/Desktop/vscode_folder]
$ export G=5

(akarsh@fearless)-[~/Desktop/vscode_folder]
$ export P=23

(akarsh@fearless)-[~/Desktop/vscode_folder]
$ python client2.py
Server IP: 192.168.0.102
[+] Connected to 192.168.0.102:5555
Username: alice
Shared Secret: qwerty
[-] AUTH FAILED
Username: alice
Shared Secret: qwerty
[-] AUTH FAILED
Username: alice
Shared Secret: qwerty
[-] AUTH FAILED
[-] Authentication failed 3 times. Connection closed.

(akarsh@fearless)-[~/Desktop/vscode_folder]
$
```

Now here is the Wireshark Captures for TASK 2

Here the capture shows the client with IP=192.168.0.105 has given its username as “alice” along with a shared secret key to the server IP=192.168.0.102 for authentication.

```
110 9.632508017 192.168.0.105 192.168.0.102 TCP 71 43986 → 5555 [PSH, ACK] Seq=1 Ack=18 Win=64512 Len=5 TSval=1113553634 TSecr=2124799403
111 9.632549171 192.168.0.102 192.168.0.105 TCP 66 5555 → 43986 [ACK] Seq=10 Ack=6 Win=65536 Len=0 TSval=2124801439 TSecr=1113553634
112 9.633264349 192.168.0.102 192.168.0.105 TCP 73 5555 → 43986 [PSH, ACK] Seq=10 Ack=6 Win=65536 Len=7 TSval=2124801440 TSecr=1113553634
113 9.674158616 192.168.0.105 192.168.0.102 TCP 66 43986 → 5555 [ACK] Seq=6 Ack=17 Win=64512 Len=0 TSval=1113553795 TSecr=2124801440
129 13.316728865 192.168.0.105 192.168.0.102 TCP 130 43986 → 5555 [PSH, ACK] Seq=6 Ack=17 Win=64512 Len=64 TSval=1113557343 TSecr=2124801440

Frame 110: Packet, 71 bytes on wire (568 bits), 71 bytes captured (568 bits) on interface wlan0, id 0
  Ethernet II, Src: AzureWaveTec_1a:a7:25 (34:6f:24:1a:a7:25), Dst: Intel_06:03:e2 (ac:12:03:06:03:e2)
  Internet Protocol Version 4, Src: 192.168.0.105, Dst: 192.168.0.102
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 57
  Identification: 0xb03f (45119)
  010 .... = Flags: 0x2, Don't Fragment
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 64
  Protocol: TCP (6)
  Header Checksum: 0x0800 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 192.168.0.105
  Destination Address: 192.168.0.102
  [Stream index: 9]

0000 ac 12 03 06 03 e2 34 6f 24 1a a7 25 08 00 45 00 .....4o $ % E
0010 00 39 b0 3f 40 00 40 06 08 06 c9 a8 00 69 c0 a8 :9 0 0 : .....f
0020 00 66 ab 02 15 b3 ec 59 50 df e1 db 3e 0c 08 00 18 :f : > : YP :
0030 00 3f 8c 5b 00 00 01 01 08 0a 42 5f 7a e2 7e a5 ? [ : : : _B_z :
0040 dd e7 61 6c 69 03 65 :alice
```

This capture shows the client with IP=192.168.0.105 has generated in the form as HASH (nonce || shared_secret) and give it to server for authentication .

```
112 9.633264349 192.168.0.102 192.168.0.105 TCP 73 5555 → 43986 [PSH, ACK] Seq=10 Ack=6 Win=65536 Len=7 TSval=2124801440 TSecr=1113553634
113 9.674158616 192.168.0.105 192.168.0.102 TCP 66 43986 → 5555 [ACK] Seq=6 Ack=17 Win=64512 Len=0 TSval=1113553795 TSecr=2124801440
129 13.316728865 192.168.0.105 192.168.0.102 TCP 130 43986 → 5555 [PSH, ACK] Seq=6 Ack=17 Win=64512 Len=64 TSval=1113557343 TSecr=2124801440
130 13.317369951 192.168.0.102 192.168.0.105 TCP 78 5555 → 43986 [PSH, ACK] Seq=17 Ack=70 Win=65536 Len=12 TSval=2124805124 TSecr=1113557343
131 13.521572362 192.168.0.105 192.168.0.102 TCP 66 43986 → 5555 [ACK] Seq=70 Ack=29 Win=64512 Len=0 TSval=1113557548 TSecr=2124805124
132 13.521687313 192.168.0.102 192.168.0.105 TCP 67 5555 → 43986 [PSH, ACK] Seq=29 Ack=70 Win=65536 Len=1 TSval=2124805328 TSecr=1113557548 [Malformed Packet]

Frame 112: Packet, 73 bytes on wire (584 bits), 73 bytes captured (584 bits) on interface wlan0, id 0
  Ethernet II, Src: Intel_06:03:e2 (ac:12:03:06:03:e2), Dst: AzureWaveTec_1a:a7:25 (34:6f:24:1a:a7:25)
  Internet Protocol Version 4, Src: 192.168.0.102, Dst: 192.168.0.105
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 59
  Identification: 0xeed4 (61140)
  010 .... = Flags: 0x2, Don't Fragment
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 64
  Protocol: TCP (6)
  Header Checksum: 0xc9c8 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 192.168.0.102
  Destination Address: 192.168.0.105
  [Stream index: 9]

0000 34 6f 24 1a a7 25 ac 12 03 06 03 e2 08 00 45 00 4o$ % : .....E
0010 00 3b ee d4 40 00 40 06 c9 c8 c0 a8 00 66 c0 a8 :0 0 0 : .....f
0020 00 69 15 b3 ab 82 e1 db 3e 0c ec 59 50 e4 80 18 :1 : > : YP :
0030 00 40 82 4d 00 00 01 01 08 0a 7e a5 e5 a0 42 5f :@ R : : : _B_
0040 7a e2 36 39 34 32 34 39 0a :z 694249 :
```

In this capture it shows that the server (192.168.0.102) after checking the HASH generated by client gives a response as “AUTH SUCCESS” which tells that client is authenticated by server.

```
130 13.317369951 192.168.0.102 192.168.0.105 TCP 78 5555 → 43986 [PSH, ACK] Seq=17 Ack=70 Win=65536 Len=12 TSval=2124805124 TSecr=1113557343
131 13.521572362 192.168.0.105 192.168.0.102 TCP 66 43986 → 5555 [ACK] Seq=70 Ack=29 Win=64512 Len=0 TSval=1113557548 TSecr=2124805124
132 13.521687313 192.168.0.102 192.168.0.105 TCP 67 5555 → 43986 [PSH, ACK] Seq=29 Ack=70 Win=65536 Len=1 TSval=2124805328 TSecr=1113557548 [Malformed Packet]

Frame 130: Packet, 78 bytes on wire (624 bits), 78 bytes captured (624 bits) on interface wlan0, id 0
  Ethernet II, Src: Intel_06:03:e2 (ac:12:03:06:03:e2), Dst: AzureWaveTec_1a:a7:25 (34:6f:24:1a:a7:25)
  Internet Protocol Version 4, Src: 192.168.0.102, Dst: 192.168.0.105
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 64
  Identification: 0xeed5 (61141)
  010 .... = Flags: 0x2, Don't Fragment
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 64
  Protocol: TCP (6)
  Header Checksum: 0xc9c2 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 192.168.0.102
  Destination Address: 192.168.0.105
  [Stream index: 9]

0000 34 6f 24 1a a7 25 ac 12 03 06 03 e2 08 00 45 00 4o$ % : .....E
0010 00 40 ee d5 40 00 40 06 c9 c2 c0 a8 00 66 c0 a8 :0 0 0 : .....f
0020 00 69 15 b3 ab 82 e1 db 3e 13 ec 59 51 24 80 18 :1 : > : YQ$ :
0030 00 40 82 52 08 00 01 01 08 0a 7e a5 f4 04 42 5f :@ R : : : _B_
0040 89 5f 41 55 54 20 53 55 43 43 45 53 53 :_AUTH S UCCSS
```


This consecutive capture shows that server(192.168.0.102) is showing the client and server public key value as 2 and 21 respectively is generated for session establishment.

132	13.521607313	192.168.0.102	192.168.0.105	TCP	67	5555 → 43906 [PSH, ACK] Seq=29 Ack=70 Win=65536 Len=1 TSval=2124805328 TSecr=1113557548 [Malformed Packet]
133	13.727662902	192.168.0.105	192.168.0.102	TCP	66	43906 → 5555 [ACK] Seq=70 Ack=30 Win=64512 Len=0 TSval=1113557753 TSecr=2124805328
134	13.727663522	192.168.0.105	192.168.0.102	TCP	68	43906 → 5555 [PSH, ACK] Seq=70 Ack=30 Win=64512 Len=2 TSval=1113557753 TSecr=2124805328
135	13.768776521	192.168.0.102	192.168.0.105	TCP	66	5555 → 43906 [ACK] Seq=30 Ack=72 Win=65536 Len=0 TSval=2124805576 TSecr=1113557753

Frame 132: Packet, 67 bytes on wire (536 bits), 67 bytes captured (536 bits) on interface wlan0, id 0
 Ethernet II, Src: Intel_06:03:e2 (ac:12:03:06:03:e2), Dst: AzureWaveTec_1a:a7:25 (34:6f:24:1a:a7:25)
 Internet Protocol Version 4, Src: 192.168.0.102, Dst: 192.168.0.105
 0100 = Version: 4
 0101 = Header Length: 20 bytes (5)
 0000 34 6f 24 1a a7 25 ac 12 03 06 03 e2 00 00 45 00 40 \$ % % % % E
 0010 00 35 ee d6 40 00 40 06 c9 cc c0 a8 00 06 c0 a8 5 @ @ % % % f
 0020 00 69 15 b3 ab 82 e1 db 3e 1f ec 59 51 24 00 18 i % % % % % Y Q \$ %
 0030 00 40 82 47 00 00 01 01 08 0a 7e a5 f4 d0 42 5f 2
 0040 8a 2a 32 2

No.	Time	Source	Destination	Protocol	Length	Info
129	13.316728065	192.168.0.105	192.168.0.102	TCP	138	43906 → 5555 [PSH, ACK] Seq=6 Ack=17 Win=64512 Len=64 TSval=1113557343 TSecr=2124801440
130	13.317369951	192.168.0.102	192.168.0.105	TCP	78	5555 → 43906 [PSH, ACK] Seq=17 Ack=70 Win=65536 Len=12 TSval=2124805124 TSecr=1113557343
131	13.521572362	192.168.0.105	192.168.0.102	TCP	66	43906 → 5555 [ACK] Seq=70 Ack=29 Win=64512 Len=0 TSval=1113557548 TSecr=2124805124
132	13.521607313	192.168.0.102	192.168.0.105	TCP	67	5555 → 43906 [PSH, ACK] Seq=29 Ack=70 Win=65536 Len=1 TSval=2124805328 TSecr=1113557548 [Malformed Packet]
133	13.727662902	192.168.0.105	192.168.0.102	TCP	66	43906 → 5555 [ACK] Seq=70 Ack=30 Win=64512 Len=0 TSval=1113557753 TSecr=2124805328
134	13.727663522	192.168.0.105	192.168.0.102	TCP	68	43906 → 5555 [PSH, ACK] Seq=70 Ack=30 Win=64512 Len=2 TSval=1113557753 TSecr=2124805328
135	13.768776521	192.168.0.102	192.168.0.105	TCP	66	5555 → 43906 [ACK] Seq=30 Ack=72 Win=65536 Len=0 TSval=2124805576 TSecr=1113557753

Frame 134: Packet, 68 bytes on wire (544 bits), 68 bytes captured (544 bits) on interface wlan0, id 0
 Ethernet II, Src: AzureWaveTec_1a:a7:25 (34:6f:24:1a:a7:25), Dst: Intel_06:03:e2 (ac:12:03:06:03:e2)
 Internet Protocol Version 4, Src: 192.168.0.105, Dst: 192.168.0.102
 0100 = Version: 4
 0101 = Header Length: 20 bytes (5)
 0000 ac 12 03 06 03 e2 34 6f 24 1a a7 25 00 00 45 00 40 \$ % % % % E
 0010 00 36 b0 44 40 00 40 06 08 5e c0 a8 00 09 c0 a8 6 D @ @ % % % i
 0020 00 66 ab 82 15 b3 ec 59 51 24 e1 db 3e 20 00 18 f % % % % % Y Q \$ %
 0030 00 3f 62 a4 00 00 01 01 08 0a 42 5f 8a f9 7e a5 7b % % % % % B %
 0040 f4 00 32 31 21

Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 Total Length: 56
 Identification: 0xb045 (45125)
 010. = Flags: 0x2, Don't fragment
 ... 0 0000 0000 0000 = Fragment Offset: 0
 Time to Live: 64
 Protocol: TCP (6)
 Header Checksum: 0x085b [validation disabled]
 [Header checksum status: Unverified]
 Source Address: 192.168.0.105
 Destination Address: 192.168.0.102
 [Stream Index: 9]

This capture shows that client (192.168.0.105) has requested a **LIST** command to know the list of directory from server(192.168.0.102).

155	17.1089954753	192.168.0.105	192.168.0.102	TCP	70	43906 → 5555 [PSH, ACK] Seq=72 Ack=30 Win=64512 Len=4 TSval=1113561179 TSecr=2124805576
157	17.107008036	192.168.0.102	192.168.0.105	TCP	66	5555 → 43906 [ACK] Seq=30 Ack=76 Win=65536 Len=0 TSval=2124808914 TSecr=1113561179
159	17.107955064	192.168.0.102	192.168.0.105	TCP	111	5555 → 43906 [PSH, ACK] Seq=30 Ack=76 Win=65536 Len=45 TSval=2124808915 TSecr=1113561179
161	17.414219341	192.168.0.105	192.168.0.102	TCP	66	43906 → 5555 [ACK] Seq=76 Ack=75 Win=64512 Len=0 TSval=1113561480 TSecr=2124808915
183	24.069080556	192.168.0.105	192.168.0.102	TCP	83	43906 → 5555 [PSH, ACK] Seq=76 Ack=75 Win=64512 Len=17 TSval=1113560874 TSecr=2124808915
184	24.070633105	192.168.0.102	192.168.0.105	TCP	76	5555 → 43906 [PSH, ACK] Seq=75 Ack=93 Win=65536 Len=0 TSval=2124815877 TSecr=1113560874

Frame 155: Packet, 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface wlan0, id 0
 Ethernet II, Src: AzureWaveTec_1a:a7:25 (34:6f:24:1a:a7:25), Dst: Intel_06:03:e2 (ac:12:03:06:03:e2)
 Internet Protocol Version 4, Src: 192.168.0.105, Dst: 192.168.0.102
 0100 = Version: 4
 0101 = Header Length: 20 bytes (5)
 Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 Total Length: 56
 Identification: 0xb045 (45125)
 010. = Flags: 0x2, Don't fragment
 ... 0 0000 0000 0000 = Fragment Offset: 0
 Time to Live: 64
 Protocol: TCP (6)
 Header Checksum: 0x085b [validation disabled]
 [Header checksum status: Unverified]
 Source Address: 192.168.0.105
 Destination Address: 192.168.0.102
 [Stream Index: 9]

0000 ac 12 03 06 03 e2 34 6f 24 1a a7 25 00 00 45 00 40 \$ % % % % E
 0010 00 38 b0 45 40 00 40 06 08 5b c0 a8 00 09 c0 a8 8 @ @ % % % i
 0020 00 66 ab 82 15 b3 ec 59 51 26 e1 db 3e 20 00 18 f % % % % % Y Q \$ %
 0030 00 3f e6 09 00 00 01 01 08 0a 42 5f 98 5b 7e a5 7b % % % % % B %
 0040 f5 c8 4c 49 53 54 LIST

This capture shows that server(192.168.0.102) has given a response to client (192.168.0.105) for the LIST command as **RESPONSE: file2.txt,file1.txt, readme.md,document.txt**. This tells that the directory has these four files contained in its directory.

159	17.107955064	192.168.0.102	192.168.0.105	TCP	1115555 - 43906 [PSH, ACK] Seq=30 Ack=76 Win=65536 Len=43 TSval=2124808915 TSecr=1113561179	
161	17.414219341	192.168.0.105	192.168.0.102	TCP	6643906 - 5555 [ACK] Seq=76 Ack=75 Win=64512 Len=0 TSval=1113561480 TSecr=2124808915	
183	24.069808556	192.168.0.105	192.168.0.102	TCP	8343906 - 5555 [PSH, ACK] Seq=76 Ack=75 Win=64512 Len=17 TSval=1113560874 TSecr=2124808915	
184	24.070333105	192.168.0.102	192.168.0.105	TCP	765555 - 43906 [PSH, ACK] Seq=85 Ack=93 Win=65536 Len=10 TSval=2124815877 TSecr=1113560874	
Frame 159: Packet, 111 bytes on wire (888 bits), 111 bytes captured (888 bits) on interface wlan0, id 0						0000 34 6f 24 1a a7 25 ac 12 03 06 03 e2 08 00 45 00E-
Ethernet II, Src: Intel_06:03:e2 (ac:12:03:06:03:e2), Dst: AzureWaveTec_1a:a7:25 (34:6f:24:1a:a7:25)						0010 00 61 00 40 00 40 06 c9 9d c0 a8 00 66 c0 a8f..
Internet Protocol Version 4, Src: 192.168.0.102, Dst: 192.168.0.105						0020 00 69 15 b3 ab 82 e1 db 3e 20 ec 59 51 2a 80 18YQ>M..
0100 = Version: 4						0030 00 40 82 73 00 00 01 01 08 0a 7e a6 02 d3 42 5fB..
.... 0101 = Header Length: 20 bytes (5)						0040 98 5b 66 69 6c 65 32 2e 74 78 74 2c 20 66 69 6c [file2.txt, fil
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)						0050 65 31 2e 74 78 74 2c 20 72 65 61 64 6d 65 2e 6d e1.txt, readme.m
Total Length: 97						0060 64 2c 20 64 6f 63 75 6d 65 6e 74 2e 74 78 74d, docum ent.txt
Identification: 0xeed9 (61145)						
010. = Flags: 0x2, Don't fragment						
...0 0000 0000 0000 = Fragment Offset: 0						
Time to Live: 64						
Protocol: TCP (6)						
Header Checksum: 0xc99d [validation disabled]						
[Header checksum status: Unverified]						
Source Address: 192.168.0.102						
Destination Address: 192.168.0.105						
[Stream index: 9]						

This capture shows that client (192.168.0.105) has requested GETSIZE file1.txt command to know the size of file1.txt from server(192.168.0.102) in the form of bytes.

183	24.069808556	192.168.0.105	192.168.0.102	TCP	8343906 - 5555 [PSH, ACK] Seq=76 Ack=75 Win=64512 Len=17 TSval=1113560874 TSecr=2124808915	
184	24.070633105	192.168.0.102	192.168.0.105	TCP	765555 - 43906 [PSH, ACK] Seq=75 Ack=93 Win=65536 Len=10 TSval=2124815877 TSecr=1113560874	
191	24.175473960	192.168.0.105	192.168.0.102	TCP	6643906 - 5555 [ACK] Seq=93 Ack=85 Win=64512 Len=0 TSval=1113568300 TSecr=2124815877	
250	32.036847926	192.168.0.105	192.168.0.102	TCP	8343906 - 5555 [PSH, ACK] Seq=93 Ack=85 Win=64512 Len=17 TSval=1113576150 TSecr=2124815877	
251	32.036392216	192.168.0.102	192.168.0.105	TCP	765555 - 43906 [PSH, ACK] Seq=85 Ack=95 Win=65536 Len=10 TSval=2124823843 TSecr=1113576150	
Frame 183: Packet, 83 bytes on wire (664 bits), 83 bytes captured (664 bits) on interface wlan0, id 0						0000 ac 12 03 06 03 e2 34 6f 24 1a a7 25 08 00 45 004o\$ % -E-
Ethernet II, Src: AzureWaveTec_1a:a7:25 (34:6f:24:1a:a7:25), Dst: Intel_06:03:e2 (ac:12:03:06:03:e2)						0010 00 45 b0 47 40 00 40 06 08 4c c0 a8 00 69 c0 a8E6-@-L...i..
Internet Protocol Version 4, Src: 192.168.0.105, Dst: 192.168.0.102						0020 00 66 ab 82 15 b3 ec 59 51 2a e1 db 3e 4d 80 18f.....YQ'>M..
0100 = Version: 4						0030 00 3f 47 b6 00 00 01 01 08 0a 42 5f b3 4a 7e a6?G.....B..J-
.... 0101 = Header Length: 20 bytes (5)						0040 02 d3 47 45 54 53 49 5a 45 20 66 69 6c 65 31 2eGETSIZ E file1.
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)						0050 74 78 74txt
Total Length: 69						
Identification: 0xb047 (45127)						
010. = Flags: 0x2, Don't fragment						
...0 0000 0000 0000 = Fragment Offset: 0						
Time to Live: 64						
Protocol: TCP (6)						
Header Checksum: 0xb04c [validation disabled]						
[Header checksum status: Unverified]						
Source Address: 192.168.0.105						
Destination Address: 192.168.0.102						
[Stream index: 9]						

This capture shows that server(192.168.0.102) has given response to client (192.168.0.105) for the GETSIZE file1.txt command as **6395 bytes** which signify that complete file with name file1.txt has 6395 bytes of data in it.

184	24.070633105	192.168.0.102	192.168.0.105	TCP	765555 - 43906 [PSH, ACK] Seq=75 Ack=93 Win=65536 Len=10 TSval=2124815877 TSecr=1113560874	
191	24.175473960	192.168.0.105	192.168.0.102	TCP	6643906 - 5555 [ACK] Seq=93 Ack=85 Win=64512 Len=0 TSval=1113568300 TSecr=2124815877	
250	32.036847926	192.168.0.105	192.168.0.102	TCP	8343906 - 5555 [PSH, ACK] Seq=93 Ack=85 Win=64512 Len=17 TSval=1113576150 TSecr=2124815877	
251	32.036392216	192.168.0.102	192.168.0.105	TCP	765555 - 43906 [PSH, ACK] Seq=85 Ack=95 Win=65536 Len=10 TSval=2124823843 TSecr=1113576150	
252	32.043694754	192.168.0.105	192.168.0.102	TCP	6643906 - 5555 [ACK] Seq=110 Ack=95 Win=64512 Len=0 TSval=1113576165 TSecr=2124823843	
361	41.785438824	192.168.0.105	192.168.0.102	TCP	8043906 - 5555 [PSH, ACK] Seq=110 Ack=95 Win=64512 Len=14 TSval=1113585820 TSecr=2124823843	
192	17.070333105	192.168.0.102	192.168.0.105	TCP	1025555 - 43906 [PSH, ACK] Seq=95 Ack=74 Win=65536 Len=0 TSval=2124815877 TSecr=1113560874	
Frame 184: Packet, 76 bytes on wire (608 bits), 76 bytes captured (608 bits) on interface wlan0, id 0						0000 34 6f 24 1a a7 25 ac 12 03 06 03 e2 08 00 45 00E-
Ethernet II, Src: Intel_06:03:e2 (ac:12:03:06:03:e2), Dst: AzureWaveTec_1a:a7:25 (34:6f:24:1a:a7:25)						0010 00 3e b0 48 40 00 40 06 c9 bf c0 a8 00 66 c0 a8>@-@-.....f..
Internet Protocol Version 4, Src: 192.168.0.102, Dst: 192.168.0.105						0020 00 60 15 b3 ab 82 e1 db 3e 4d ec 59 51 3b 80 181.....>M YQ:..
0100 = Version: 4						0030 00 40 82 50 00 00 01 01 08 0a 7e a6 1e 05 42 5f@P.....B..
.... 0101 = Header Length: 20 bytes (5)						0040 b3 4a 36 33 39 35 29 62 79 74 65 73J6395 b ytes
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)						
Total Length: 62						
Identification: 0xb048 (45146)						
010. = Flags: 0x2, Don't fragment						
...0 0000 0000 0000 = Fragment Offset: 0						
Time to Live: 64						
Protocol: TCP (6)						
Header Checksum: 0xc9bf [validation disabled]						
[Header checksum status: Unverified]						
Source Address: 192.168.0.102						
Destination Address: 192.168.0.105						
[Stream index: 9]						


```

361 41.785438824 192.168.0.105 192.168.0.102 TCP 80 43906 - 5555 [PSH, ACK] Seq=110 Ack=95 Win=64512 Len=14 TSval=1113585820 TSecr=2124823843
362 41.786372194 192.168.0.102 192.168.0.105 TCP 169 5555 - 43906 [PSH, ACK] Seq=95 Ack=124 Win=65536 Len=103 TSval=2124833593 TSecr=1113585820
363 41.993613771 192.168.0.105 192.168.0.102 TCP 80 43906 - 5555 [ACK] Seq=124 Ack=198 Win=64512 Len=0 TSval=1113586028 TSecr=2124833593
483 48.792266415 192.168.0.105 192.168.0.102 TCP 80 43906 - 5555 [PSH, ACK] Seq=124 Ack=198 Win=64512 Len=14 TSval=1113592913 TSecr=2124833593
494 48.792947707 192.168.0.102 192.168.0.105 TCP 169 5555 - 43906 [PSH, ACK] Seq=198 Ack=138 Win=65536 Len=103 TSval=2124840060 TSecr=1113592913

> Frame 361: Packet, 80 bytes on wire (640 bits), 80 bytes captured (640 bits) on interface wlan0, id 0
> Ethernet II, Src: AzureWaveTc1a:a7:25 (34:f6:24:a1:a7:25), Dst: Intel_06:03:e2 (ac:12:03:06:03:e2)
> Internet Protocol Version 4, Src: 192.168.0.105, Dst: 192.168.0.102
  0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total Length: 66
  Identification: 0xb04c (45132)
  010. .... = Flags: 0x2, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 64
  Protocol: TCP (6)
  Header Checksum: 0xb04a [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 192.168.0.105
  Destination Address: 192.168.0.102
  [Stream index: 0]
0000 ac 12 03 06 03 e2 34 f6 24 1a a7 25 08 00 45 00 ... 40 $ % - E -
0010 00 42 b0 4c 48 08 48 06 08 4a c0 a8 09 69 c0 a8 ... B 10 0 . J . . 1 .
0020 00 66 ab 82 15 b3 ec 59 51 4c e1 b0 3e 61 80 18 ... f . . . Y Q L > a .
0030 00 3f b8 df 00 00 01 01 08 0a 42 5f f8 9c 7e a6 ... ? . . . . B . . .
0040 3d 23 49 4e 46 4f 20 66 69 6c 65 31 2e 74 78 74 ... =INFO f ile1.txt

```

```

1 362 41.786372194 192.168.0.102 192.168.0.105 TCP 169 5555 - 43906 [PSH, ACK] Seq=95 Ack=124 Win=65536 Len=103 TSval=2124835993 TSecr=1113585820
363 41.9393613771 192.168.0.105 192.168.0.102 TCP 66 43906 - 5555 [ACK] Seq=124 Ack=198 Win=64512 Len=0 TSval=1113586020 TSecr=2124833593
403 48.792286415 192.168.0.105 192.168.0.102 TCP 80 43906 - 5555 [PSH, ACK] Seq=124 Ack=198 Win=64512 Len=14 TSval=1113592913 TSecr=2124833593
494 48.792954787 192.168.0.102 192.168.0.105 TCP 169 5555 - 43906 [PSH, ACK] Seq=198 Ack=138 Win=55536 Len=103 TSval=2124840000 TSecr=1113592913

Frame 362: Packet, 169 bytes on wire (1352 bits), 169 bytes captured (1352 bits) on interface wlan0, id 0
Ethernet II, Src: Intel_06:03:e2 (ac:12:03:06:03:e2), Dst: AzureWaveTec_1a:a7:25 (34:6f:24:1a:a7:25)
Internet Protocol Version 4, Src: 192.168.0.102, Dst: 192.168.0.105
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 155
Identification: 0xeedc (61148)
010. .... = Flags: 0x2, Don't fragment
... 0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: TCP (6)
Header Checksum: 0xc906 [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.0.102
Destination Address: 192.168.0.105
[Stream index: 9]
0000 34 6f 24 1a a7 25 ac 12 03 06 03 e2 08 00 45 00 40s % .....E..
0010 00 0b 95 4d 40 00 48 06 c9 60 c0 08 06 c0 a8 .....@-...f..
0020 00 0b 15 b3 ab 82 e1 db 3e 61 ec 59 51 5a 80 18 .....a YQZ..
0030 00 00 48 ac 42 00 00 01 01 08 0a te a6 63 39 42 5f @.....=cB0
0040 f8 9c 4e 61 73 2a 2d 4d 6f 64 69 66 69 65 64 3a ..Last Modified:
0050 2d 32 3e 30 2d 2d 38 32 2d 31 31 20 32 32 3a 33 ..2026-02-11 22:3
0060 39 3a 34 35 2e 36 33 35 31 35 31 2c 20 43 72 65 9:45:635 151, Cre
0070 61 74 65 64 3a 20 32 38 32 36 2d 30 32 2d 31 31 ..20:20 26:02-1
0080 2d 32 32 3a 34 30 3a 38 35 2e 37 33 34 30 36 31 ..atted: 05.734061
0090 2c 20 50 65 72 6d 69 73 7d 69 6f 6e 73 3a 20 2d ..Permissions: -
00a0 72 77 2d 72 72 72 72 2d 72 2d ..rw-rw-r--

```

```

[ *tcp.port = 5555 | ip.addr = 192.168.0.105
No.      Time      Source                Destination            Protocol  Length  Info
-----
404 48.792964707 192.168.0.102        192.168.0.105         TCP      109    5555 -> 43906 [PSH, ACK] Seq=198 Ack=138 Win=65536 Len=103 TSval=2124840600 TSecr=1113592913
405 48.797406995 192.168.0.105        192.168.0.102         TCP      66    43906 -> 5555 [ACK] Seq=138 Ack=301 Win=64512 Len=0 TSval=1113592922 TSecr=2124840600
520 55.984745110 192.168.0.102        192.168.0.102         TCP      70    43906 -> 5555 [PSH, ACK] Seq=138 Ack=301 Win=64512 Len=4 TSval=1113600106 TSecr=2124840600
521 55.985502786 192.168.0.102        192.168.0.105         TCP      66    5555 -> 43906 [FIN, ACK] Seq=301 Ack=142 Win=65536 Len=0 TSval=2124847792 TSecr=1113600106
522 55.993575900 192.168.0.105        192.168.0.102         TCP      66    43906 -> 5555 [FIN, ACK] Seq=142 Ack=302 Win=64512 Len=0 TSval=1113600119 TSecr=2124847792
523 55.993616725 192.168.0.102        192.168.0.105         TCP      66    5555 -> 43906 [ACK] Seq=302 Ack=143 Win=65536 Len=0 TSval=2124847800 TSecr=1113600119

> Frame 520: Packet, 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface wlan0, id 0
> Ethernet II, Src: AzureWaveTec 1a:7a:25 (34:6f:24:1a:7a:25), Dst: Intel_06:03:e2 (ac:12:03:06:03:e2)
> Internet Protocol Version 4, Src: 192.168.0.105, Dst: 192.168.0.102
  0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 56
    Identification: 0x0850 (45136)
  > 010. .... = Flags: 0x2, Don't fragment
    .... 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 64
    Protocol: TCP (6)
    Header Checksum: 0x0850 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.0.105
    Destination Address: 192.168.0.102
    Stream index: 91
  >
  0000  ac 12 03 06 03 e2 34 6f  24 1a 7a 25 08 00 45 00  .... 4o $ % E .
0010  00 30 00 00 40 00 40 00  08 50 c0 a8 09 0f c0 a8  - 0 0 0 . . . . 1 .
0020  00 60 a0 82 15 b3 ec 59  51 68 e1 d0 3f 2f 80 18  - 0 0 0 . Y . . . ? .
0030  00 3f c9 3c 00 00 01 01  08 0a 42 60 30 6a 7e a6  - ? . . . . 0 0 0 ? .
0040  7e 98 51 55 49 54                - QUIT

```

Task 3: File Download with Integrity

Summary and Design Overview

The goal of Task 3 is to ensure **Data Integrity**, while Task 2 secured the session, Task 3 ensures that the files being downloaded are not tampered with or corrupted during transition. Here the "Chunked Transfer" mechanism where each 1024-byte block of data is verified using a Message Authentication Code (MAC) and tracked with a Sequence Number to prevent out-of-order delivery.

Implementation Approach

Based on the code provided, here is the technical breakdown:

File Chunking: The server reads the requested file in fixed blocks of **1024 bytes**. If the final block is smaller, it reads only the remaining bytes.

Sequence Numbering: A variable `seq_no` (starting at 0) is incremented for every chunk. This ensures that even if two chunks contain identical data, their MACs will be different, and the order is strictly enforced.

MAC Generation: For every chunk, the server calculates:

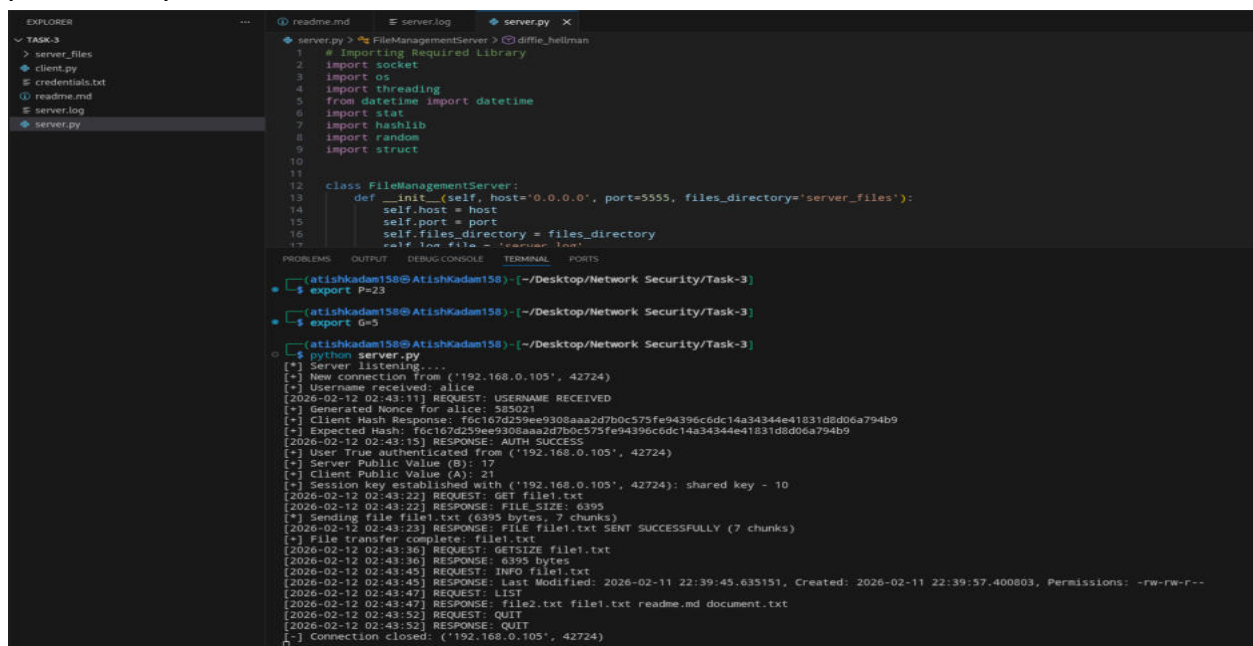
MAC = HASH(DATA || SEQ_NO || SESSION_KEY)

SERVER SIDE

This is the `server.py` file terminal capture

which shows how in TASK3 the `file1.txt` (total bytes of `file1.txt` is 6395) is transferred to client (192.168.0.102) in chunks of 1024 bytes

here in terminal capture of Server shows **"RESPONSE: FILE file1.txt SENT SUCCESSFULLY (7 CHUNKS)"** this shows the file is transferred to client.



```
server.py > FileManagementServer > diffie_hellman
1 # Importing Required Library
2 import socket
3 import os
4 import threading
5 from datetime import datetime
6 import stat
7 import hashlib
8 import random
9 import struct
10
11
12 class FileManagementServer:
13     def __init__(self, host="0.0.0.0", port=5555, files_directory="server_files"):
14         self.host = host
15         self.port = port
16         self.files_directory = files_directory
17         self.log_file = "server.log"
18
19
20 PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
21
22 [atishkadam158@AtishKadam158] ~/Desktop/Network Security/Task-3
23 $ export P=23
24 [atishkadam158@AtishKadam158] ~/Desktop/Network Security/Task-3
25 $ export G=5
26 [atishkadam158@AtishKadam158] ~/Desktop/Network Security/Task-3
27 $ python server.py
28 [*] Server listening....
29 [*] New connection from ('192.168.0.105', 42724)
30 [*] Username received: alice
31 [2026-02-12 02:43:11] REQUEST: USERNAME RECEIVED
32 [*] Generated Nonce for alice: 585021
33 [*] Client Hash Response: f6c167d259ee9308aaa2d7b0c573fe94396c6dc14a34344e41831dbd06a794b9
34 [*] Expected Hash: f6c167d259ee9308aaa2d7b0c573fe94396c6dc14a34344e41831dbd06a794b9
35 [2026-02-12 02:43:15] RESPONSE: AUTH SUCCESS
36 [*] User True authenticated from ('192.168.0.105', 42724)
37 [*] Server Public Value (B): 17
38 [*] Client Public Value (A): 21
39 [*] Session key established with ('192.168.0.105', 42724): shared key - 10
40 [2026-02-12 02:43:22] REQUEST: GET file1.txt
41 [2026-02-12 02:43:22] RESPONSE: FILE_SIZE: 6395
42 [*] Sending file file1.txt (6395 bytes, 7 chunks)
43 [2026-02-12 02:43:22] RESPONSE: FILE file1.txt SENT SUCCESSFULLY (7 chunks)
44 [*] File transfer complete: file1.txt
45 [2026-02-12 02:43:36] REQUEST: GETSIZE file1.txt
46 [2026-02-12 02:43:36] RESPONSE: 6395 bytes
47 [2026-02-12 02:43:45] REQUEST: INFO file1.txt
48 [2026-02-12 02:43:45] RESPONSE: Last Modified: 2026-02-11 22:39:45.635151, Created: 2026-02-11 22:39:57.400803, Permissions: -rw-rw-r--
49 [2026-02-12 02:43:47] REQUEST: LIST
50 [2026-02-12 02:43:47] RESPONSE: file2.txt file1.txt readme.md document.txt
51 [2026-02-12 02:43:52] REQUEST: QUIT
52 [2026-02-12 02:43:52] RESPONSE: QUIT
53 [*] Connection closed: ('192.168.0.105', 42724)
```

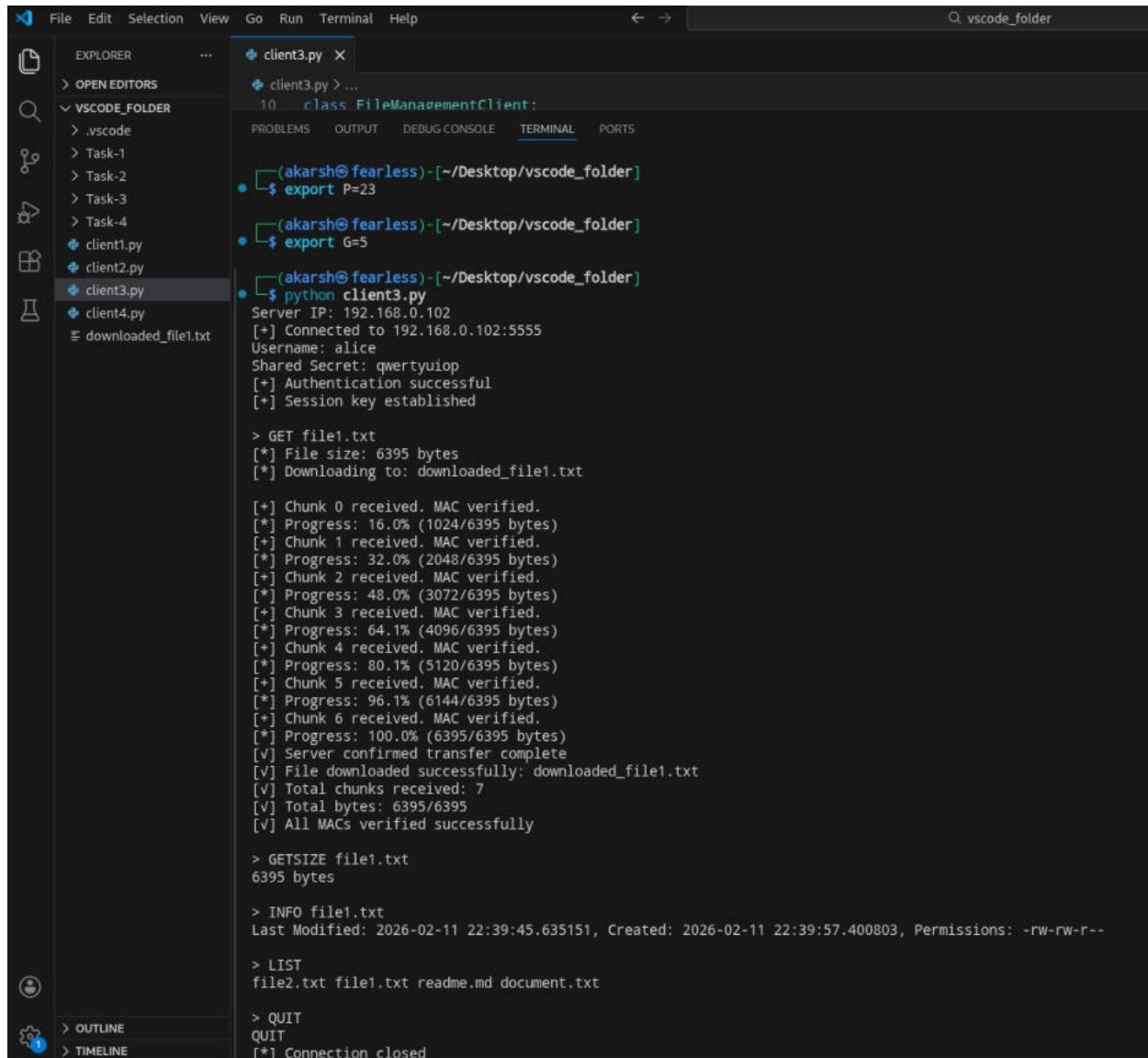

Client side

This is the [client3.py](#) file terminal capture

which shows how in TASK3 the file1.txt (total bytes of file1.txt is 6395) is received to client (192.168.0.102) in chunks of 1024 bytes

here in terminal capture of client shows

“File downloaded successfully: downloaded_file1.txt” this shows the file is transferred to client by server in the form of chunks 0 to chunk 6 along with MAC verified for each chunk.



```
File Edit Selection View Go Run Terminal Help
client3.py X
class FileManagementClient:
10
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
(akarsh@fearless)~/Desktop/vscode_folder
$ export P=23
(akarsh@fearless)~/Desktop/vscode_folder
$ export G=5
(akarsh@fearless)~/Desktop/vscode_folder
$ python client3.py
Server IP: 192.168.0.102
[+] Connected to 192.168.0.102:5555
Username: alice
Shared Secret: qwertyuiop
[+] Authentication successful
[+] Session key established

> GET file1.txt
[*] File size: 6395 bytes
[*] Downloading to: downloaded_file1.txt

[+] Chunk 0 received. MAC verified.
[*] Progress: 16.0% (1024/6395 bytes)
[+] Chunk 1 received. MAC verified.
[*] Progress: 32.0% (2048/6395 bytes)
[+] Chunk 2 received. MAC verified.
[*] Progress: 48.0% (3072/6395 bytes)
[+] Chunk 3 received. MAC verified.
[*] Progress: 64.1% (4096/6395 bytes)
[+] Chunk 4 received. MAC verified.
[*] Progress: 80.1% (5120/6395 bytes)
[+] Chunk 5 received. MAC verified.
[*] Progress: 96.1% (6144/6395 bytes)
[+] Chunk 6 received. MAC verified.
[*] Progress: 100.0% (6395/6395 bytes)
[v] Server confirmed transfer complete
[v] File downloaded successfully: downloaded_file1.txt
[v] Total chunks received: 7
[v] Total bytes: 6395/6395
[v] All MACs verified successfully

> GETSIZE file1.txt
6395 bytes

> INFO file1.txt
Last Modified: 2026-02-11 22:39:45.635151, Created: 2026-02-11 22:39:57.400803, Permissions: -rw-rw-r--

> LIST
file2.txt file1.txt readme.md document.txt

> QUIT
QUIT
[*] Connection closed
```

Now here is the wireshark capture of TASK 3

This is the wireshark capture of TASK 3 from server(192.168.0.102) to client (192.168.0.105) requesting for USERNAME from client after connected with user in order to check for valid authentication.

130	10.144476956	192.168.0.102	192.168.0.105	TCP	75 5555 → 57408 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=9 TSval=2132083987 TSecr=1120836214
131	10.344984383	192.168.0.105	192.168.0.102	TCP	66 57408 → 5555 [ACK] Seq=1 Ack=10 Win=64512 Len=0 TSval=1120836415 TSecr=2132083987
141	12.916202352	192.168.0.105	192.168.0.102	TCP	71 57408 → 5555 [PSH, ACK] Seq=1 Ack=10 Win=64512 Len=5 TSval=1120839072 TSecr=2132083987
142	12.916244443	192.168.0.102	192.168.0.105	TCP	66 5555 → 57408 [ACK] Seq=10 Ack=6 Win=65536 Len=0 TSval=2132086758 TSecr=1120839072
143	12.916911660	192.168.0.102	192.168.0.105	TCP	73 5555 → 57408 [PSH, ACK] Seq=10 Ack=6 Win=65536 Len=7 TSval=2132086759 TSecr=1120839072

Frame 130: Packet, 75 bytes on wire (600 bits), 75 bytes captured (600 bits) on interface wlan0, id 0	0000	34 6f 24 1a a7 25 ac 12 03 06 03 e2 08 00 45 00	40\$ %E
Ethernet II, Src: Intel_06:03:e2 (ac:12:03:06:03:e2), Dst: AzureWaveTec_1a:a7:25 (34:6f:24:1a:a7:25)	0010	00 3d 68 a0 40 00 40 06 4f fb c0 a8 00 66 c0 a8	=h @ @ 0 f . .
Internet Protocol Version 4, Src: 192.168.0.102, Dst: 192.168.0.105	0020	00 69 15 b3 e0 40 62 fd 5c 31 d7 69 22 c1 80 18	i . . @ . \ 1 i " . .
0100 = Version: 4	0030	00 40 82 4f 00 00 01 01 08 0a 7f 15 05 13 42 ce	@ 0B . . .
.... 0101 = Header Length: 20 bytes (5)	0040	9a 76 55 53 45 52 4e 41 4d 45 3a	vUSERNA ME:
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)			
Total Length: 61			
Identification: 0x68a0 (26784)			
010. = Flags: 0x2, Don't fragment			
...0 0000 0000 0000 = Fragment Offset: 0			
Time to Live: 64			
Protocol: TCP (6)			
Header Checksum: 0x4fffb [validation disabled]			
[Header checksum status: Unverified]			
Source Address: 192.168.0.102			
Destination Address: 192.168.0.105			
[Stream index: 8]			

This is the wireshark capture of TASK 3 from client (192.168.0.105) to server(192.168.0.102) in which the user client is giving response of username as “alice” along with shared secret for authentication from server.

141	12.916202352	192.168.0.105	192.168.0.102	TCP	71 57408 → 5555 [PSH, ACK] Seq=1 Ack=10 Win=64512 Len=5 TSval=1120839072 TSecr=2132083987
142	12.916244443	192.168.0.102	192.168.0.105	TCP	66 5555 → 57408 [ACK] Seq=10 Ack=6 Win=65536 Len=0 TSval=2132086758 TSecr=1120839072
143	12.916911660	192.168.0.102	192.168.0.105	TCP	73 5555 → 57408 [PSH, ACK] Seq=10 Ack=6 Win=65536 Len=7 TSval=2132086759 TSecr=1120839072
144	12.922976418	192.168.0.105	192.168.0.102	TCP	66 57408 → 5555 [ACK] Seq=6 Ack=17 Win=64512 Len=0 TSval=1120839091 TSecr=2132086759
164	16.697178205	192.168.0.105	192.168.0.102	TCP	130 57408 → 5555 [PSH, ACK] Seq=6 Ack=17 Win=64512 Len=64 TSval=1120842770 TSecr=2132086759
165	16.697887342	192.168.0.102	192.168.0.105	TCP	78 5555 → 57408 [PSH, ACK] Seq=17 Ack=70 Win=65536 Len=12 TSval=2132090540 TSecr=1120842770

Frame 141: Packet, 71 bytes on wire (568 bits), 71 bytes captured (568 bits) on interface wlan0, id 0	0000	ac 12 03 06 03 e2 34 6f 24 1a a7 25 08 00 45 00	40 \$ % E
Ethernet II, Src: AzureWaveTec_1a:a7:25 (34:6f:24:1a:a7:25), Dst: Intel_06:03:e2 (ac:12:03:06:03:e2)	0010	00 39 f6 fd 40 00 40 06 c1 a1 c0 a8 00 69 c0 a8	9 : @ @ i . .
Internet Protocol Version 4, Src: 192.168.0.105, Dst: 192.168.0.102	0020	00 66 e0 40 15 b3 d7 69 22 c1 62 fd 5c 3a 80 18	f @ . i " b \ : . .
0100 = Version: 4	0030	00 3f a8 93 00 00 01 01 08 0a 42 ce a5 a0 7f 15	?B
.... 0101 = Header Length: 20 bytes (5)	0040	05 13 61 6c 69 63 65	..alice
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)			
Total Length: 57			
Identification: 0xf6fd (63229)			
010. = Flags: 0x2, Don't fragment			
...0 0000 0000 0000 = Fragment Offset: 0			
Time to Live: 64			
Protocol: TCP (6)			
Header Checksum: 0xc1a1 [validation disabled]			
[Header checksum status: Unverified]			
Source Address: 192.168.0.105			
Destination Address: 192.168.0.102			

This is the wireshark capture of TASK 3 from server(192.168.0.102) to client (192.168.0.105) to tell that “AUTH SUCCESS” which means that the username credential is being authenticated by the server.

```

# tcp port 5555 | ip.addr 192.168.0.105
No.    Time           Source                Destination           Protocol Length Info
143.12.916244443  192.168.0.102        192.168.0.105        TCP                    66 5555 - 57408 [ACK] Seq=10 Ack=6 Win=65536 Len=0 TSval=52132086758 TSecr=1126839072
143.12.916916666  192.168.0.102        192.168.0.105        TCP                    73 5555 - 57408 [PSH, ACK] Seq=10 Ack=6 Win=65536 Len=7 TSval=1232086759 TSecr=1126839072
143.12.922976418  192.168.0.105        192.168.0.102        TCP                    73 57408 - 5555 [ACK] Seq=8 Ack=17 Win=64512 Len=0 TSval=1126839091 TSecr=1232086759
143.12.930717805  192.168.0.102        192.168.0.105        TCP                    130 57408 - 5555 [ACK] Seq=8 Ack=17 Win=64512 Len=0 TSval=1126842770 TSecr=1232086759
143.12.937837342  192.168.0.102        192.168.0.105        TCP                    78 5555 - 57408 [PSH, ACK] Seq=17 Ack=31 Win=65536 Len=15 TSval=1232090458 TSecr=1126842770
143.12.968463518  192.168.0.105        192.168.0.102        TCP                    67 57408 - 5555 [ACK] Seq=78 Ack=29 Win=64512 Len=0 TSval=1126842971 TSecr=1232090458
143.12.968463518  192.168.0.105        192.168.0.102        TCP                    67 5555 - 57408 [ACK] Seq=29 Ack=78 Win=65536 Len=1 TSval=1232090741 TSecr=1126842971 [Malformed Packet]
143.12.105176781  192.168.0.105        192.168.0.102        TCP                    67 57408 - 5555 [ACK] Seq=78 Ack=38 Win=64512 Len=0 TSval=1232095074 TSecr=1232090741

# Frame 105: Packet, 78 bytes on wire (624 bits), 78 bytes captured (624 bits) on interface wlan0, id 0
# Ethernet II, Src: IntelE168:03:e2 (ac:12:30:03:03:e2), Dst: AzureWaveTec:1a:a7:25 (34:6f:24:1a:a7:25)
# Internet Protocol Version 4, Src: 192.168.0.102, Dst: 192.168.0.105
0100 ... = Version: 4
... 0101 = Header Length: 20 bytes (2)
# Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 64
Identification: 0x68a3 (26787)
# 010 ... = Flags: 0x2, Don't Fragment
... 0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: TCP (6)
Header Checksum: 0xaff5 [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.0.102
Destination Address: 192.168.0.105
Stream: 1064518
AUTH SUCCESS

```

This is the wireshark capture of TASK 3 from client (192.168.0.105) to server(192.168.0.102) in which client requests to get the file1.txt from server .

```
[tcp.port == 5555] | ip.addr == 192.168.0.105
```

No.	Time	Source	Destination	Protocol	Length Info
170	21.14545769	192.168.0.192	192.168.0.105	TCP	60 5555 → 57408 [ACK] Seq=30 Ack=71 Win=65530 Len=0 TSval=2132090988 TSecr=1120493173
292	21.14545993	192.168.0.105	192.168.0.192	TCP	79 57408 → 5555 [PSH, ACK] Seq=71 Ack=30 Win=64512 Len=0 TSval=1120494278 TSecr=2132099988
293	21.14528074	192.168.0.192	192.168.0.105	TCP	66 5555 → 57408 [ACK] Seq=30 Ack=84 Win=65530 Len=0 TSval=2132096987 TSecr=112049278
294	21.145018796	192.168.0.192	192.168.0.105	TCP	88 5555 → 57408 [PSH, ACK] Seq=84 Ack=84 Win=65536 Len=14 TSval=2132096988 TSecr=112049278
295	21.14502978	192.168.0.192	192.168.0.105	TCP	14860 Connection Refused [RST] Seq=1000000000 Win=0 Len=0 TSval=1120849768 TSecr=2132096988
296	23.659256758	192.168.0.105	192.168.0.192	TCP	71 57408 → 5555 [PSH, ACK] Seq=84 Ack=44 Win=64512 Len=0 TSval=1120849796 TSecr=2132096988
297	23.6592527	192.168.0.192	192.168.0.105	TCP	78 168.0.0.0 ACK 2031 → 5555 57408 [ACK] Seq=71 Ack=84 Win=65530 Len=0 TSval=1120849791 TSecr=1120849768 SLE=71 SWE=84
298	23.65925137	192.168.0.192	192.168.0.105	TCP	65 5555 → 57408 [ACK] Seq=30 Ack=84 Win=65530 Len=0 TSval=1120849791 TSecr=1120849768

```
# Frame 202: Packet: 79 bytes on wire (632 bits), 79 bytes captured (632 bits) on interface wlan0, 16 B  
# Ethernet II, Src: AzureWaveTc...la:78:25:34;Prio:0x1e(75); Dst: Intel:le03:e2 (ac:12:93:06:93:e2)  
# Internet Protocol Version 4, Src: 192.168.0.195, Dst: 192.168.0.192  
0900 .... = Version: 4  
..... = Header Length: 20 bytes (E)  
.. Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)  
Total Length: 65  
Identification: 0xf7f3 (63295)  
09B.... = Flags: 0x02, Not: Fragment  
...0 0000 0000 0000 = Fragment Offset: 0  
Time to Live: 64  
Protocol: TCP (6)  
Header Checksum: 0xc193 [validation disabled]  
[Header checksum status: Unverified]  
Source Address: 192.168.0.195  
Destination Address: 192.168.0.192
```

```
0 40 5 5 E  
A 0 0 1  
7 0 1 6 b N  
7 0 1 6 B  
1GET f1 lel.txt
```

This is the wireshark capture of TASK 3 from server(192.168.0.102) to client (192.168.0.105) In which the server has sent the **FILESIZE of 6395** for the file1.txt to client ,later which the server will send the files in chunks to client .

```

204 23.145919796 192.168.0.102 192.168.0.105 TCP 80 5555 - 57488 [PSH, ACK] Seq=30 Ack=84 Win=65536 Len=14 TSval=2132096988 TSecr=1120849278
205 23.659255736 192.168.0.105 192.168.0.102 TCP 79 [TCP Spurious Retransmission] 57488 - 5555 [PSH, ACK] Seq=71 Ack=30 Win=64512 Len=13 TSval=1120849788 TSecr=2132099088
206 23.659256758 192.168.0.105 192.168.0.102 TCP 71 57488 - 5555 [PSH, ACK] Seq=84 Ack=44 Win=64512 Len=5 TSval=1120849796 TSecr=2132096988
207 23.659297527 192.168.0.102 192.168.0.105 TCP 78 [TCP Dup ACK 203#1] 5555 - 57488 [ACK] Seq=44 Ack=84 Win=65536 Len=8 TSval=2132097561 TSecr=1120849788 SLE=71 SRE=84
208 23.659330522 192.168.0.102 192.168.0.105 TCP 66 5555 - 57488 [ACK] Seq=44 Ack=89 Win=65536 Len=9 TSval=2132097561 TSecr=1120849796

Frame 204: Packet, 80 bytes on wire (640 bits), 80 bytes captured (640 bits) on interface wlan0, id 0
Ethernet II, Src: Intel_06:03:e2 (ac:12:03:06:03:e2), Dst: AzureWaveTec_1a:a7:25 (24:6f:24:1a:a7:25)
Internet Protocol Version 4, Src: 192.168.0.102, Dst: 192.168.0.105
  0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 66
    Identification: 0x68a7 (26791)
  010. .... = Flags: 0x2, Don't Fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 64
    Protocol: TCP (6)
    Header Checksum: 0x4fef [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.0.102
    Destination Address: 192.168.0.105
    [Stream index: 8]

```


[illegible]

```

210 23 76800194 192.168.0.105 192.168.0.102 TCP 77 57408 - 5555 [PSH, ACK] Seq=80 Ack=1104 Win=7656 Len=5 TVal=112084960 TSrc=2132097542
211 23 76800192 192.168.0.102 192.168.0.105 TCP 1126 5555 - 57408 [PSH, ACK] Seq=1104 Ack=94 Win=65536 Len=5 TVal=1120977542 TSrc=1120849606
212 23 768065392 192.168.0.102 192.168.0.102 TCP 77 57408 - 5555 [PSH, ACK] Seq=84 Ack=2164 Win=76656 Len=5 TVal=1120849628 TSrc=2132097542
213 23 768963807 192.168.0.102 192.168.0.105 TCP 1126 5555 - 57408 [PSH, ACK] Seq=2164 Ack=99 Win=65536 Len=1860 TVal=11232097603 TSrc=1128849928
214 23 768902742 192.168.0.102 192.168.0.105 TCP 5299 Ack=3724 Win=71680 Len=1 TVal=1120849937 TSrc=2132097603

+ Frame 210: Packed, 74 bytes on wire (568 bits), 71 bytes captured (568 bits) on interface wlan0, id 0
+ Ethernet II, Src: AzureWaveTec_1a:7:25 (34:f6:24:1a:7:25), Dst: Intel_06:03:e2 (ac:12:03:06:03:e2)
- Destination: Intel_06:03:e2 (ac:12:03:06:03:e2)
+ . . . . . = 4.0 bit: Globally unique address (factory default)
+ . . . . . = 10 bit: Individual address (unicast)
- Source: AzureWaveTec_1a:7:25 (34:f6:24:1a:7:25)
+ . . . . . = 4.0 bit: Globally unique address (factory default)
+ . . . . . = 10 bit: Individual address (unicast)

Type: IPv4 (0x0800)
(Stream index: 2)
+ Internet Protocol Version 4, Src: 192.168.0.105, Dst: 192.168.0.102
0100 .... = Version: 4
...0101 = Header Length: 20 bytes (5)
+ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 57
Identification: 0xf706 (63238)
0100 .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: TCP (6)
Header Checksum: 0xc198 [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.0.105
Destination Address: 192.168.0.102
(Stream index: 3)
+ Transmission Control Protocol, Src Port: 57408, Dst Port: 5555, Seq: 89, Ack: 1104, Len: 5
+ Data (5 bytes)
Length: 4143b3a30
[Length: 5]

```

Now here is the last chunk wireshark capture of chunk 6 sent by server to client.

[illegible]

```

tcp.port == 5555 || ip.addr == 192.168.0.105
No.      Time      Source      Destination      Protocol  Length  Info
219 23.779561266 192.168.0.102 192.168.0.105 TCP 1120 5555 -> 57408 [PSH, ACK] Seq=5344 Ack=114 Win=65536 Len=1060 TSval=2132097621 TSecr=1126849947
220 23.870118637 192.168.0.105 192.168.0.102 TCP 71 57408 -> 5555 [PSH, ACK] Seq=114 Ack=6484 Win=71680 Len=5 TSval=1126850143 TSecr=2132097621
224 23.979362923 192.168.0.102 192.168.0.105 TCP 353 5555 -> 57408 [PSH, ACK] Seq=6484 Ack=119 Win=65536 Len=287 TSval=2132097821 TSecr=1126850143
225 24.374353884 192.168.0.105 192.168.0.102 TCP 71 57408 -> 5555 [PSH, ACK] Seq=119 Ack=6691 Win=71680 Len=5 TSval=1126850443 TSecr=2132097621
229 24.374579153 192.168.0.102 192.168.0.105 TCP 83 5555 -> 57408 [PSH, ACK] Seq=6691 Ack=124 Win=65536 Len=17 TSval=2132098217 TSecr=1126850443
230 24.575182979 192.168.0.105 192.168.0.102 TCP 66 57408 -> 5555 [ACK] Seq=124 Ack=6708 Win=71680 Len=0 TSval=1126850609 TSecr=2132098217
358 31.790329212 192.168.0.105 192.168.0.102 TCP 83 57408 -> 5555 [PSH, ACK] Seq=124 Ack=6708 Win=71680 Len=17 TSval=1126857948 TSecr=2132098217
359 31.791255635 192.168.0.102 192.168.0.105 TCP 76 5555 -> 57408 [PSH, ACK] Seq=6788 Ack=141 Win=65536 Len=10 TSval=2132105633 TSecr=1126857948

+ Frame 225: Packet, 71 bytes on wire (568 bits), 71 bytes captured (568 bits) on interface wlan0, id 0
+ Ethernet II, Src: AzureWaveTec_1a:a7:25 (34:f4:24:1a:a7:25), Dst: Intel_06:03:e2 (ac:12:03:06:03:e2)
+ Destination: Intel_06:03:e2 (ac:12:03:06:03:e2)
+ .0 ..... = LG bit: Globally unique address (factory default)
+ .....0 ..... = IG bit: Individual address (unicast)
+ Source: AzureWaveTec_1a:a7:25 (34:f4:24:1a:a7:25)
+ .....0 ..... = LG bit: Globally unique address (factory default)
+ .....0 ..... = IG bit: Individual address (unicast)
+ Type: IPv4 (0x0800)
+ [Stream index: 2]
+ Internet Protocol Version 4, Src: 192.168.0.105, Dst: 192.168.0.102
+ 0100 .... = Version: 4
+ .....0101 = Header Length: 20 bytes (5)
+ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-Ect)
+ Total Length: 57
+ Identification: 0xf79c (63244)
+ 010. .... = Flags: 0x2, Don't fragment
+ ...0 0000 0000 0000 = Fragment Offset: 0
+ Time to Live: 64
+ Protocol: TCP (6)
+ Header Checksum: 0xc192 [validation disabled]
+ [Header checksum status: Unverified]
+ Source Address: 192.168.0.105
+ Destination Address: 192.168.0.102
+ [Stream index: 3]
+ Transmission Control Protocol, Src Port: 57408, Dst Port: 5555, Seq: 119, Ack: 6691, Len: 5
+ Data (5 bytes)
+ Data: 41434b3a36
+ [Length: 5]

```

```

tcp.port == 5555 || ip.addr == 192.168.0.105

No.      Time           Source                Destination           Protocol  Length  Info
1 225.24.37453884  192.168.0.105        192.168.0.102        TCP        78      57408 → 5555 [PSH, ACK] Seq=113 Ack=6691 Win=71600 Len=5 Tsval=1120856443 Tsecr=2132067821
2 225.24.37453933  192.168.0.102        192.168.0.105        TCP        78      5555 → 57408 [ACK] Seq=6691 Win=0 Len=0 Tsval=1120856443 Tsecr=2132067821
3 235.24.575102978 192.168.0.105        192.168.0.102        TCP        66      57408 → 5555 [ACK] Seq=124 Ack=6708 Win=71688 Len=0 Tsval=1120856668 Tsecr=2132098617
4 358.31.798529212 192.168.0.105        192.168.0.102        TCP        83      57408 → 5555 [PSH, ACK] Seq=124 Ack=6708 Win=71688 Len=1 Tsval=1120857948 Tsecr=2132086617
5 358.31.791250835 192.168.0.102        192.168.0.105        TCP        78      5555 → 57408 [PSH, ACK] Seq=6708 Ack=141 Win=65536 Len=0 Tsval=1121056633 Tsecr=1120857948

Frame 225: Packet, 83 bytes on wire (664 bits), 83 bytes captured (664 bits) on interface wlan0, 1d 0
  Ethernet II, Src: Intel_06:03:e2 (ac:12:03:06:03:e2), Dst: AzureWaveTec_1a:a7:25 (34:6f:24:1a:a7:25)
    Destination: AzureWaveTec_1a:a7:25 (34:6f:24:1a:a7:25)
      . . . . . = 10 bit: Globally unique address (factory default)
      . . . . . = 10 bit: Individual address (unicast)
    - Source: Intel_06:03:e2 (ac:12:03:06:03:e2)
      . . . . . = 10 bit: Globally unique address (factory default)
      . . . . . = 10 bit: Individual address (unicast)
    Type: IPv4 (0x0800)
    (Stream index: 2)
  Internet Protocol Version 4, Src: 192.168.0.102, Dst: 192.168.0.105
    @100 . . . . = Version: 4
    . . . . @101 = Header Length: 20 bytes (5)
    . Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 69
    Identification: 0x60b1 (246801)
    @10 . . . . = Flags: 0x2, Don't Fragment
    . . . @0000 @0000 @0000 = Fragment Offset: 0
    Time to Live: 64
    Protocol: TCP (6)
    Header Checksum: 0x4fc5 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.0.102
    Destination Address: 192.168.0.105
    (Stream index: 8)
  Transmission Control Protocol, Src Port: 5555, Dst Port: 57408, Seq: 6691, Ack: 124, Len: 1
    Data [16 bytes]
    Data: 45244545e334645525434af4d504455445
    [Length: 17]

```

```

358 31.796329212 192.168.0.105 192.168.0.102 TCP 83 57408 - 5555 [PSH, ACK] Seq=124 Ack=6708 Win=71680 Len=17 Tsval=1120857948 TSecr=2132086217
359 31.791256535 192.168.0.102 192.168.0.105 TCP 76 5555 - 57408 [PSH, ACK] Seq=6708 Ack=141 Win=65536 Len=10 Tsval=2132105633 TSecr=1120857948
360 31.787895218 192.168.0.105 192.168.0.102 TCP 86 57408 - 5555 [ACK] Seq=141 Ack=6718 Win=71680 Len=0 Tsval=1120857964 TSecr=2132105033
435 38.3829295605 192.168.0.105 192.168.0.102 TCP 86 57408 - 57408 [PSH, ACK] Seq=141 Ack=6718 Win=71680 Len=14 Tsval=11208604457 TSecr=2132105633
436 38.384021752 192.168.0.102 192.168.0.105 TCP 169 5555 - 57408 [PSH, ACK] Seq=6718 Ack=Win=65536 Len=103 Tsval=2132112146 TSecr=21120864457

+ Frame 358: Packet, 83 bytes on wire (664 bits), 83 bytes captured (664 bits) on interface wlan0, id 0
- Ethernet II, Src: AzureWaveTec_1a:a7:25 (34:f6:23:1a:a7:25), Dst: Intel_06:03:e2 (ac:12:03:06:03:e2)
- Destination Intel_06:03:e2 (ac:12:03:06:03:e2)
- Source AzureWaveTec_1a:a7:25 (34:f6:23:1a:a7:25)
- Length: 83
- Protocol: TCP
- Window size bytes: 65536
- Sequence number: 141 (window=65536)
- Flags: 0x00000000 = 0
- Identification: 0x7f0e (63246)
- Offset: 0
- Total length: 83
- Time to Live: 64
- Protocol: TCP (6)
- Header checksum: bc18d4 [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.0.105
Destination Address: 192.168.0.102

```



```

[ *tcp.port=5555 | ip.addr=192.168.0.105]
No.      Time            Source                Destination           Protocol Length Info
-----
358 31.796329212 192.168.0.105        192.168.0.102        TCP                    83 57488 -> 5555 [PSH, ACK] Seq=124 Ack=6766 Win=71680 Len=17 TSval=1120857948 TSecr=2132098621
359 31.796329212 192.168.0.105        192.168.0.102        UDP                    128 5555->5555 [ACK] Seq=141 Ack=6718 Win=71680 Len=0 TSval=1120857964 TSecr=2132105633
360 31.797095276 192.168.0.105        192.168.0.102        TCP                    66 57488 -> 5555 [ACK] Seq=141 Ack=6718 Win=71680 Len=0 TSval=1120857964 TSecr=2132105633
361 38.302995605 192.168.0.105        192.168.0.102        TCP                    80 57488 -> 5555 [PSH, ACK] Seq=141 Ack=6718 Win=71680 Len=14 TSval=1120864457 TSecr=2132105633
362 38.304821752 192.168.0.105        192.168.0.102        TCP                    169 5555 -> 57488 [PSH, ACK] Seq=8718 Ack=155 Win=65536 Len=103 TSval=2132112146 TSecr=1120864457

+ Frame 359: Packet, 76 bytes on wire (608 bits), 76 bytes captured (608 bits) on interface wlan0, id 0
- Ethernet II, Src: Intel_06:03:e2 (ac:12:03:06:03:e2), Dst: AzureWaveTec_1a:a7:25 (34:f6:24:1a:a7:25)
  - Destination: AzureWaveTec_1a:a7:25 (34:f6:24:1a:a7:25)
    ....0..... = IG bit: Globally unique address (factory default)
    .....0..... = IG bit: Individual address (unicast)
  - Source: Intel_06:03:e2 (ac:12:03:06:03:e2)
    ....0..... = IG bit: Globally unique address (factory default)
    .....0..... = IG bit: Individual address (unicast)

Type: IPv4 (0x0000)
[Stream index: 2]
- Internet Protocol Version 4, Src: 192.168.0.102, Dst: 192.168.0.105
  0100 .... = Version: 4
  ....0101 = Header Length: 20 bytes (5)
  - Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 62
    Identification: 0xbdb2 (28802)
    010. .... = Flags: 0x2, Don't fragment
    ...0000 0000 0000 = Fragment Offset: 0
    Time to Live: 64
    Protocol: TCP (6)
    Header Checksum: 0x4feb [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.0.102
    Destination Address: 192.168.0.105
    [Stream index: 0]

```

```
tcp.port == 5555 || ip.addr == 192.168.0.105
```

No.	Time	Source	Destination	Protocol	Length	Info
482	44.854323679	192.168.0.105	192.168.0.102	TCP	78	57408 → 5555 [PSH, ACK] Seq=159 Ack=6863 Win=71680 Len=4 TSval=1120876960 TSecr=2132115934
483	44.854323679	192.168.0.105	192.168.0.102	TCP	10	5555 → 57408 [RST, ACK] Seq=6863 Win=0 Len=0 TSval=1120876960 TSecr=1120876960
484	44.855116209	192.168.0.102	192.168.0.105	TCP	66	5555 → 57408 [FIN, ACK] Seq=6867 Ack=163 Win=55536 Len=0 TSval=2132115937 TSecr=1120876960
487	45.061410696	192.168.0.105	192.168.0.102	TCP	66	57408 → 5555 [ACK] Seq=163 Ack=6867 Win=71680 Len=0 TSval=1120871127 TSecr=2132118097
488	45.061411299	192.168.0.105	192.168.0.102	TCP	66	57408 → 5555 [FIN, ACK] Seq=163 Ack=6868 Win=71680 Len=0 TSval=1120871127 TSecr=2132118097

```
Frame 483: Packet, 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface wlan0, id 0
  Ethernet II, Src: Intel_06:03:e2 (ac:12:03:06:03:e2), Dst: AzureWaveTec_1a:a7:25 (34:6f:24:1a:a7:25)
    Destination: AzureWaveTec_1a:a7:25 (34:6f:24:1a:a7:25)
      0000 00 34 6f 24 1a a7 25 = IG bit: Globally unique address (factory default)
      0010 00 38 68 b5 08 00 40 86 = IG bit: Individual address (unicast)
      0020 00 40 42 84 00 00 01 01 00 0a 7f 15 0c a9 42 cf = IG bit: Globally unique address (factory default)
      0030 00 40 42 84 00 00 01 01 00 0a 7f 15 0c a9 42 cf = IG bit: Individual address (unicast)
      0040 22 30 51 55 49 54
    Type: IPv4 (0x0800)
    [Stream index: 2]
  Internet Protocol Version 4, Src: 192.168.0.102, Dst: 192.168.0.105
    0100 ..... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    + Differentiated Services Field: 0x000 (DSCP: CS0, ECN: Not-ECT)
      Total Length: 56
      Identification: 0x08b5 (26805)
      0100 ..... = Flags: 0x2, Don't Fragment
      ... 0000 0000 0000 = Fragment Offset: 0
      Time to Live: 64
      Protocol: TCP (6)
      Header Checksum: 0x4feb [validation disabled]
      [Header checksum status: Unverified]
      Source Address: 192.168.0.102
      Destination Address: 192.168.0.105
      [Stream index: 8]
  Transmission Control Protocol, Src Port: 5555, Dst Port: 57408, Seq: 6863, Ack: 163, Len: 4
  Data (4 bytes)
```

Task 4: Encryption of Communication

Summary and Design Overview

For Task 4, the main goal was to move from just checking for tampering (integrity) to full-blown privacy (confidentiality). In the previous tasks, we made sure the session was safe, but here we're making sure that even if someone sniffs the network while `file1.txt` is being sent, they can't read a single word of it. I implemented **Authenticated Encryption**, which basically means every 1024-byte chunk is locked up tight using the AES key we generated during the handshake. This keeps the file private while also keeping the order-tracking from the previous tasks so the data doesn't get scrambled.

Implementation Approach

Looking at the terminal logs, you can see the process handled a 4988-byte file by breaking it into 5 chunks. Here's how I put it together:

- **Setting up the Keys:** First off, the client and server use that "qwertyuiop" shared secret to spin up a **Session Key and an AES Key**. Without this successful auth, the file transfer won't even start.
- **Chunk-by-Chunk Encryption:** I set the server to read the file in 1024-byte blocks. Before sending, each block is run through AES-256. In the screenshots, you can see the huge strings of hex code labeled **[Cipher]**—that's the raw encrypted data for things like Chunk 0 and Chunk 3.
- **On-the-fly Decryption:** Once the client gets a chunk, it uses the local AES key to decrypt it back into **[Plain]** text immediately. You can see the actual text about "Network Security" popping up in the logs as it decodes.
- **Double-Checking Metadata:** Before the download even begins, the client runs **GETSIZE** to confirm the file is exactly 4988 bytes. I also used an **INFO** command to pull the file permissions (`-rw-rw-r--`) and timestamps to make sure we're getting the right version of the file.
- **Final Verification:** The client tracks the count (Chunk 0 to 4) and only gives the **[v] Transfer Complete** signal once the total chunk count matches what was expected from the **GETSIZE** command.

Server Side

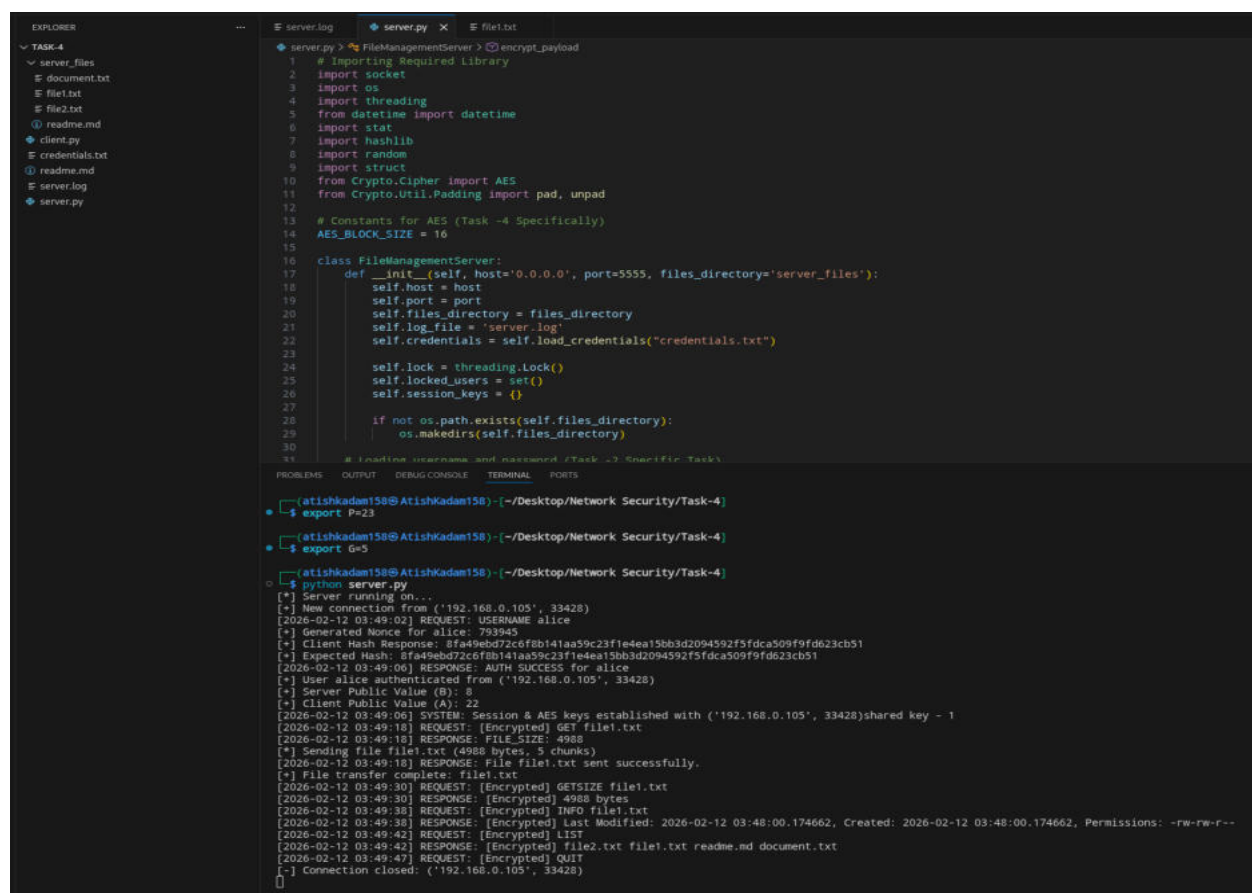
The server starts by exporting the necessary DH parameters (P=23,G=5) to set up the secure channel. Once the client connects, the server handles the handshake, verifying the username "alice" and the shared secret "qwertyuiop" to establish the AES session keys.

Command Processing: The images show the server responding to specific control commands like GET, GETSIZE, INFO, and LIST. For the GET command.

The Encryption: This is the main part of Task 4 is that the server reads file1.txt in chunks, and for each one, it generates that long string of "Cipher" text. It's essentially taking the plain text and encrypting it into a format that's unreadable without the session key before sending it.

Metadata Handling: When the client asks for INFO or LIST, the server looks up the file system details (like the -rw-rw-r-- permissions and timestamps) and sends those back so the client can verify the file's state on the server side.

Graceful Exit: Finally, when the server receives the QUIT command, it closes the individual session and logs the "Connection closed" message, showing a clean end to the secure transaction.



The screenshot displays a code editor with a file explorer on the left and a terminal window at the bottom. The file explorer shows a project named 'TASK-4' containing files like 'server_files', 'document.txt', 'file1.txt', 'file2.txt', 'readme.md', 'client.py', 'credentials.txt', 'server.log', and 'server.py'. The main editor window shows the code for 'server.py', which is a Python script for a file management server. The code includes imports for 'os', 'socket', 'threading', 'datetime', 'stat', 'hashlib', 'random', 'struct', 'Crypto.Cipher', and 'Crypto.Util.Padding'. It defines a 'FileManagementServer' class with methods for handling connections, authentication, and file operations. The terminal window shows the execution of the server and the client's interactions. The client sets the DH parameters (P=23, G=5) and then connects to the server. The server logs the connection and the client's requests (GET, GETSIZE, INFO, LIST, QUIT). The server responds with the requested data, including the encrypted file content and file metadata.

```
server.py > FileManagementServer > encrypt_payload
1 # Importing Required Library
2 import socket
3 import os
4 import threading
5 from datetime import datetime
6 import stat
7 import hashlib
8 import random
9 import struct
10 from Crypto.Cipher import AES
11 from Crypto.Util.Padding import pad, unpad
12
13 # Constants for AES (Task -4 Specifically)
14 AES_BLOCK_SIZE = 16
15
16 class FileManagementServer:
17     def __init__(self, host='0.0.0.0', port=5555, files_directory='server_files'):
18         self.host = host
19         self.port = port
20         self.files_directory = files_directory
21         self.log_file = 'server.log'
22         self.credentials = self.load_credentials('credentials.txt')
23
24         self.lock = threading.Lock()
25         self.locked_users = set()
26         self.session_keys = {}
27
28         if not os.path.exists(self.files_directory):
29             os.makedirs(self.files_directory)
30
31 # Inactive username and password (Task -3 Specific Task)
```

```
atishkadam158@AtishKadam158:~/Desktop/Network Security/Task-4
$ export P=23
$ export G=5
$ python server.py
[*] Server running on...
[*] New connection from ('192.168.0.105', 33428)
[2026-02-12 03:49:02] REQUEST: USERNAME alice
[*] Generated Nonce for alice: 793945
[*] Client Hash Response: 8fa49ebd72c6f8b141aa59c23f1e4ea15bb3d2094592f5fdca509f9fd623cb51
[*] Expected Hash: 8fa49ebd72c6f8b141aa59c23f1e4ea15bb3d2094592f5fdca509f9fd623cb51
[2026-02-12 03:49:06] RESPONSE: AUTH SUCCESS for alice
[*] User alice authenticated from ('192.168.0.105', 33428)
[*] Server Public Value (G): 5
[*] Client Public Value (A): 22
[2026-02-12 03:49:06] SYSTEM: Session & AES keys established with ('192.168.0.105', 33428)shared key - 1
[2026-02-12 03:49:18] REQUEST: [Encrypted] GET file1.txt
[2026-02-12 03:49:18] RESPONSE: [Encrypted] FILE_SIZE: 4988
[*] Sending file file1.txt (4988 bytes, 5 chunks)
[2026-02-12 03:49:18] RESPONSE: File file1.txt sent successfully.
[*] File transfer complete: file1.txt
[2026-02-12 03:49:30] REQUEST: [Encrypted] GETSIZE file1.txt
[2026-02-12 03:49:30] RESPONSE: [Encrypted] 4988 bytes
[2026-02-12 03:49:38] REQUEST: [Encrypted] INFO file1.txt
[2026-02-12 03:49:38] RESPONSE: [Encrypted] Last Modified: 2026-02-12 03:48:00.174662, Created: 2026-02-12 03:48:00.174662, Permissions: -rw-rw-r--
[2026-02-12 03:49:42] REQUEST: [Encrypted] LIST
[2026-02-12 03:49:42] RESPONSE: [Encrypted] file2.txt file1.txt readme.md document.txt
[2026-02-12 03:49:47] REQUEST: [Encrypted] QUIT
[*] Connection closed: ('192.168.0.105', 33428)
```

```
[*] [akashsh fearless] (~/.Desktop/vscode_folder)
[*] export Pw3
[*] akashsh fearless! (~/.Desktop/vscode_folder)
[*] export Gm5
[*] akashsh fearless! (~/.Desktop/vscode_folder)
[*] python client.py
[*] Connected to 192.168.0.102
[*] Connected to 192.168.0.102:5555
Username: alice
Password: qwertyuiop
[*] Authentication successful
[*] Session key & AES key established

> GET file.txt
[*] File Size: 4088 bytes
[*] Downloading to: downloaded_file.txt
-----
[*] Chunk 0:

[Cipher]: fe9f8e4ad6c9af168336d3ca2f272f480178104351f85b4662c0c7e55efbf291bf1a2ff67a7bb0b097c84ebcc5f16fd9fa4999feaf11dab3dbda9bfafeddc593702d28c4465aa4384dc3ff4db9fed10f54edd48163c33df787672eb3defbe
...
[Plain]: ---- This Sample file for user -----
What Is Network Security?
Network security protects networks and the data they carry from unauthorized access, misuse, and cyberattacks. It ensures systems remain confidential, available, and trustworthy across all digital environments.
Prevents unauthorized access and cyber threats
Protects data integrity, confidentiality, and availability
Uses layered security controls across devices, users, and systems
Ensures safe communication and reliable network operations
Network Security Goals
How Does Network Security Work?
Network security works through multiple protective layers that control access at both the network edge and inside the environment. Each layer ensures only authorized users can access resources while blocking threats. The core idea is to protect large amounts of data using layered defenses that enforce rules before any action is allowed. These layers include:
Physical security prevention
Physical security prevents
[Problem]: Unauthorized individuals from physically accessing hardware, servers, or devices.
Uses access cards, biometrics, CCTV, and locked server rooms
Protects routers, switches, cables, and data centers
Prevents tampering, device theft, and hardware-layer attacks
Technical Network Security
Technical security protects data as it is stored, processed, and transmitted across the network using software-based controls.
Includes firewalls, encryption, antivirus, and intrusion detection
Administrative Network Security
Administrative security defines policies, user permissions, and procedures that control how the network is accessed and managed.
Enforces security policies and access controls
Manages proper configuration, monitoring, and compliance
Types of Network Security
There are two major types of network security controls that protect networks from breaches, unauthorized
```


Now this is the Wireshark for TASK 4

This capture shows the initial TCP three-way handshake and the start of the application-level communication between the client (192.168.0.102) and the server (192.168.0.105).

At this stage, the connection is established, and the client is identifying itself to the server.

```

168 7.934470819 192.168.0.105 192.168.0.102 TCP 66 47258 - 5555 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=1125164826 TSrc=2136412377
169 7.93542762 192.168.0.102 192.168.0.102 TCP 75 5555 - 47258 [PSH, ACK] Seq=1 Ack=1 Win=65536 Len=9 TSval=112516412593 TSrc=1125164826
170 7.944992507 192.168.0.105 192.168.0.102 TCP 66 47258 - 5555 [ACK] Seq=1 Ack=10 Win=64512 Len=0 TSval=1125164826 TSrc=2136412503
171 7.95027290905 192.168.0.102 192.168.0.102 TCP 71 47258 - 5555 [PSH, ACK] Seq=1 Ack=10 Win=64512 Len=5 TSval=1125169824 TSrc=2136412593
172 7.9530771906 192.168.0.102 192.168.0.105 TCP 66 5555 - 47258 [ACK] Seq=10 Ack=6 Win=65536 Len=0 TSval=2136419630 TSrc=2125169824
173 7.9534060958 192.168.0.102 192.168.0.105 TCP 72 5555 - 47258 [PSH, ACK] Seq=10 Ack=6 Win=65536 Len=6 TSval=2136419631 TSrc=2125169824
174 7.9536926213 192.168.0.105 192.168.0.102 TCP 66 47258 - 5555 [ACK] Seq=6 Seq=10 Win=64512 Len=0 TSval=1125171952 TSrc=2136419631

Frame 1680: Packet, 75 bytes on wire (600 bits), 75 bytes captured (600 bits) on interface wlan0, id 0
Ethernet II, Src: Intel_06:03:e2 (ac:12:03:06:03:e2), Dst: AzureWaveTc_1a:a7:25 (34:6f:24:1a:a7:25)
Internet Protocol Version 4, Src: 192.168.0.102, Dst: 192.168.0.105
0100 .... = Version: 4
.... 0101 .. = Header Length: 20 bytes (5C)
0 Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 61
Identification: 0x441f (17439)
010, .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: TCP (6)
Header Checksum: 0x747c [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.0.102
Destination Address: 192.168.0.105
[Stream 10]hex: 111

```

This capture indicates that the client is pushing the first set of data, which includes the **Username** as **"alice"**. for authentication

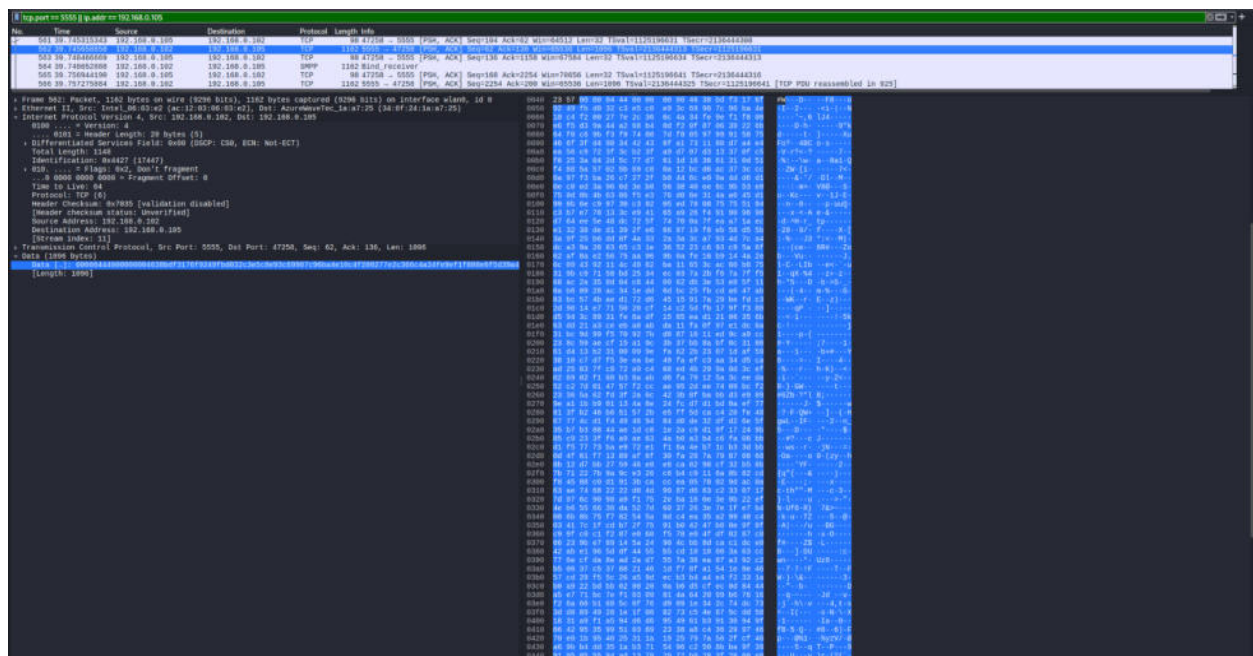
[illegible]

This snippet captures the critical moment of authentication and session key establishment. The hexadecimal data on the right decodes to **"AUTH SUCCESS"**. This confirms that the server has verified the shared secret and both parties have now derived the AES session keys needed for Task 4.

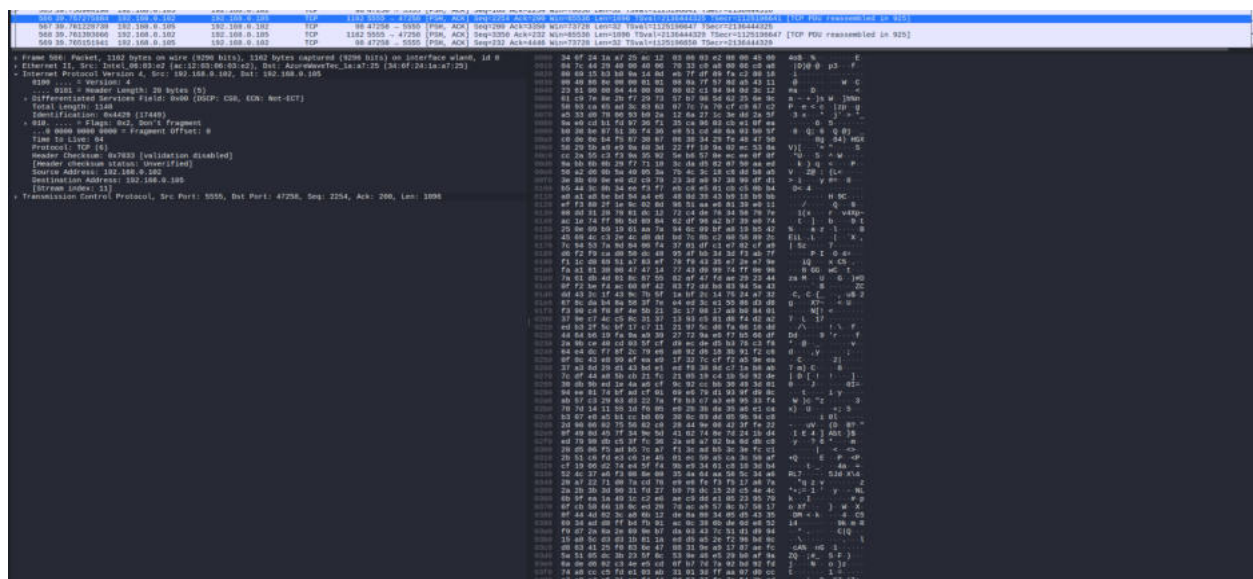
[illegible]

In this Wireshark, this is a clear view of an active encrypted data transfer for **Chunk 0**. Unlike the previous authentication messages, the "Data" field here is completely unreadable in the ASCII preview.

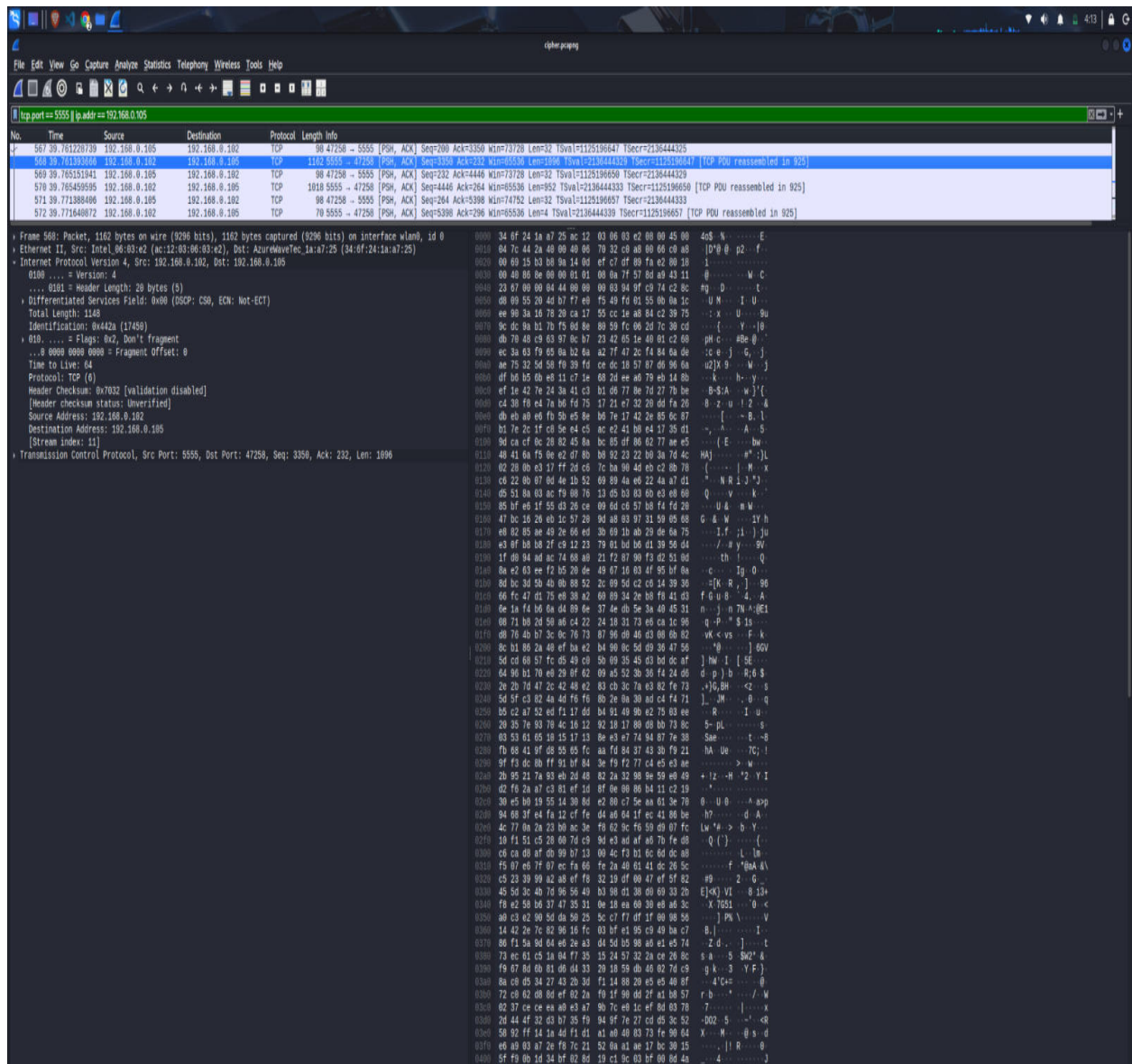
This is the **[Cipher]** text in action; it proves that even though Wireshark can see the packet, it cannot reveal the contents of the file being downloaded.



Similarly this is the clear view of an active encrypted data transfer for **Chunk 1** of file1.txt. The **[Cipher]** text in action proves that even though Wireshark can see the packet, it cannot reveal the contents of the file being downloaded for chunk 1.



Similarly this is the clear view of an active encrypted data transfer for chunk 2 of file1.txt. The **[Cipher]** text in action proves that even though Wireshark can see the packet, it cannot reveal the contents of the file being downloaded for chunk 2



So in this way this last capture of a chunk of file1.txt shows the tail end of the file transfer eventual termination of the session.

The final data segments are sent before the client issues the QUIT command seen in the terminal images. The network traffic reflects the completion of the 4988-byte file transfer across multiple reassembled TCP segments.

576.39.19545995	192.168.0.102	192.168.0.105	TCP	1935.5555 - 47250 [PSH, ACK] Seq=4448 Ack=5398 Win=65536 Len=952 TSval=1125196657 TSecr=1125196650 [TCP PDU reassembled in 925]
571.39.77180406	192.168.0.105	192.168.0.102	TCP	80.47258 - 5555 [PSH, ACK] Seq=264 Ack=5398 Win=74752 Len=32 TSval=1125196657 TSecr=2130444333
572.39.77180407	192.168.0.102	192.168.0.105	TCP	70.5555 - 47250 [PSH, ACK] Seq=5398 Ack=290 Win=65536 Len=0 TSval=1125196657 TSecr=1125196657 [TCP PDU reassembled in 925]
573.39.821617423	192.168.0.105	192.168.0.102	TCP	80.47258 - 5555 [ACK] Seq=290 Ack=5402 Win=74752 Len=0 TSval=1125196708 TSecr=2130444333
852.51.33303954	192.168.0.105	192.168.0.102	TCP	114.47258 - 5555 [PSH, ACK] Seq=290 Ack=5402 Win=74752 Len=48 TSval=1125200039 TSecr=2130444339
853.51.334251838	192.168.0.102	192.168.0.105	TCP	80.5555 - 47250 [PSH, ACK] Seq=5402 Ack=3444 Win=65536 Len=32 TSval=1125200039 TSecr=1125200039 [TCP PDU reassembled in 925]

Frame 578: Packet, 3018 bytes on wire (8144 bits), 3018 bytes captured (8144 bits) on interface wlan0, id 0

Ethernet II, Src: Intel_06:03:e2 (ac:12:83:06:03:e2), Dst: AzureWaveTec_1a:a7:25 (34:0f:24:1a:a7:25)

Internet Protocol Version 4, Src: 192.168.0.102, Dst: 192.168.0.105

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 1904

Identification: 0x442b (17451)

0100 = Flags: 0x2, Don't fragment

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 64

Protocol: TCP (6)

Header checksum: 0x79c1 [validation disabled]

[Header checksum status: Unverified]

Source Address: 192.168.0.102

Destination Address: 192.168.0.105

[Stream index: 11]

Transmission Control Protocol, Src Port: 5555, Dst Port: 47250, Seq: 4448, Ack: 264, Len: 952

0000 34 0f 24 1a a7 25 ac 12 03 06 03 e2 06 00 45 06 408 NE
0010 03 ec 44 2b 40 00 40 06 70 c1 c0 a8 00 46 c0 a5 048 0 p...f...
0020 06 09 15 b3 b0 0a 14 0d f4 0f df 09 f0 02 80 19 1.....
0030 00 40 05 fe 00 00 01 01 00 0a 7f 57 0d ad 43 11 0.....M.C
0040 23 6a 00 00 03 b4 00 00 00 04 ac f6 0b e0 ac c9 0.....L...
0050 0a ba 53 e3 74 20 b2 e4 70 e7 e1 df 0b 4e 0e 50 S 1]P
0060 73 51 20 a1 7b 29 78 c0 df 44 fe 67 47 2c 09 40 0.....
0070 17 f1 33 ad 4d 33 03 29 40 f2 64 7a 0e 00 e1 63 3 H (.....C
0080 d2 8a bd ba 5e cb e2 bd 01 c2 c7 14 df 34 03 4404 D
0090 c0 3c 03 c3 03 c0 2d 9d 41 14 1f 50 e4 39 97 37 <.....A.V 9-7
00a0 42 2a cf e0 fe aa 73 71 f2 33 30 1a c3 92 42 b130
00b0 e1 60 60 58 61 00 30 ad 74 90 25 32 b2 8e 55 e50 1 32 U
00c0 0a 0b ff 03 61 07 37 0a 11 2d 3f 9c f8 08 09 d3a 7 -7
00d0 09 0a 04 ae 3e 4c 0e c3 93 b0 4d f2 51 89 72 78L.....M.Q.rv
00e0 19 50 0c 1b bd a1 b4 35 06 03 47 ab 0b 30 a7 b9].....5 6 k;...
00f0 0f 52 02 04 20 c1 3f 60 c9 ab b3 fc 4d 02 f0 44e..4a7h.....W 3
0100 a5 60 3f 50 0b e5 e0 90 90 9f 63 a5 fc 42 53 377k.....c..S7
0110 00 3f ac 57 32 b4 e0 78 a9 c1 09 0e cc ce e0 727 k2.....v.....f
0120 98 e3 ac 60 ac 57 1f bf 94 e2 2d ab cc 7e 61 31f.W.....a1
0130 14 29 3b e0 76 a3 5d a4 0c ab 05 cc 00 11 72 44].....v.....fD
0140 0e aa 61 0f 32 10 f1 00 06 f5 25 f1 c7 c3 202 6N 9
0150 e5 5f 23 c0 06 7f db d5 e3 f6 ba f9 aa 00 9e 918.....
0160 17 35 b2 0b 00 03 03 07 f6 70 aa bd bd 34 9a 01 S.....c.....p=4 a
0170 ca e0 02 0d cf 10 bf ba 25 a6 0e 3b 45 07 9e 12b.....N;E.....
0180 40 c7 fe f0 40 05 92 55 99 90 13 ee 43 94 93 00.....U.....C.....
0190 2f ad 10 ab 3a 5c aa ec d1 ca 00 4e 06 5d 2d fef.....N.....N].....
01a0 3a 58 e0 93 09 01 21 0d 60 2c 1f 0f 00 00 aa b0X.....1 m.....V.....
01b0 45 7f ce 0f 70 e0 7e 07 e0 e5 61 2b a6 4c 9ak.....p.....g.....a+L.....
01c0 00 57 31 17 f0 75 1e 94 ad a5 a9 fb aa ad 8c 49M.....U.....I.....
01d0 65 ad 2a 7b 59 2c 09 4c de c0 71 a6 c0 0e b0 9fe.....(V.....L.....q.....m.....
01e0 c4 0b be 07 0b 2b e1 05 55 f6 b3 42 02 be 0e 0e0.....U.....B.....L.....
01f0 f4 0b 1f 79 0f b2 e0 79 29 f6 27 db 40 02 90 b4y.....y.....].....0.....
0200 5d ec f2 f3 71 02 76 b6 49 3b 4a cb fa 28 0a].....0q.....V.....I.....J.....(
0210 15 32 0d cb c0 07 45 01 98 07 d4 17 15 1e bd 132.....Ea.....
0220 76 03 25 90 b3 c3 12 b6 2d 06 05 90 4d 48 0fv.....N.....(0F.....
0230 2f 33 52 fb a5 a5 03 f6 02 b0 00 3e 03 37 edZ.....S.....b.....>Y7.....
0240 7a 0b 53 e0 e0 21 4f 7b 09 ce aa 25 3a 4e 09 91<S.....10.....[.....K.....F.....
0250 40 37 3e 04 e9 2b 71 93 40 ea b7 c0 c0 c2 777.....+.....q.....0.....w.....
0260 bf 09 4b 01 cc a1 ac a1 c0 ec 59 30 05 ee e2 50K.....K.....Y.....B.....].....
0270 30 1a f7 ee 50 0b 2a fa 32 04 43 79 70 44 9a 3e>.....X.....*.....2.....Cyd.....>.....
0280 a5 72 f5 14 50 a7 7a 71 61 f2 64 a7 03 40 10r.....0.....z.....qa.....0.....0.....
0290 f7 0b 05 c0 f0 30 e0 7f 25 b0 7a 3c 50 03 b6 c90.....0.....%.....z.....[.....
02a0 3e 0e 04 34 b5 3e 5c 7e ef 09 59 47 02 ea 1a fb>.....4.....>.....N.....Y.....G.....>.....
02b0 92 37 f2 72 ba d0 b3 12 20 9e 47 30 03 00 5c b27.....>.....&.....G.....0.....\.....
02c0 e2 aa d3 14 46 13 20 79 05 ef c9 52 30 1c b4 0cF.....ky.....R.....0.....
02d0 dc 0c b6 27 04 53 09 94 03 ac fa 57 16 0e f3 332.....[.....c.....].....pd.....b.....L.....
02e0 f2 b3 aa 07 09 40 20 72 b2 b1 ab 7a 0d 02 47 b0T.....].....>.....2.....G.....
02f0 51 4d 0d 32 0e bc 1c bf 00 7b a3 c8 05 b0 30 4cQ.....2.....>.....{.....U.....BL.....
0300 c0 ab 25 e0 11 a4 03 05 a0 d6 4f be e5 05 94 ceN.....c.....0.....>.....
0310 f2 e0 04 d0 0c 0b 4b ed 7c 7a 0c a4 04 2e 95 a5>.....K.....[.....z.....>.....
0320 dc 0c b6 27 04 53 09 94 03 ac fa 57 16 0e f3 33>.....f.....S.....G.....W.....0.....
0330 00 60 b0 42 0a 0f 0f ad 93 0d da 04 5d 4e b0 3ff.....B.....>.....TA.....?.....
0340 5d 02 4b 53 b0 34 0a ac 20 af c9 49 c0 0d 10 20].....0K.....4.....{.....I.....+.....
0350 00 ad 08 d0 d0 11 b0 09 51 71 ac de 0e fd 40 f0>.....>.....Qa.....>.....0.....
0360 f7 7f fb a4 d0 0b c3 ad f2 39 0e 00 0c 44 55>.....k.....>.....9.....K.....LDU.....
0370 24 19 ee ff 70 36 40 9c 9c a3 14 31 af 09 b0 09S.....>.....[01.....>.....I.....>.....
0380 00 65 e0 3c 04 5d 17 0b ac 1c ee 07 0e b0 00 79>.....e.....<.....].....>.....y.....
0390 0e 9d 3c 3a 0a 35 03 07 c3 9f 07 bd 09 00 ba b1<.....S.....>.....>.....
03a0 ef c7 39 52 e9 c0 09 be 14 e0 0d 0f f7 40 b0 31>.....9R.....1.....m.....F.....1.....
03b0 90 9a 5f 74 97 09 c1 e1 50 bf bc 7b 08 de 04 b3>.....t.....L.....P.....{.....>.....
03c0 a2 c0 20 b0 72 04 1c 4b 4d 0a b0 b0 f1 75 e0>.....>.....f.....K.....M.....>.....u.....
03d0 81 30 ab f3 33 00 00 9c e0 4e 24 04 0f 0e c0 c3>.....0.....>.....D.....>.....NB.....>.....
03e0 12 25 7b f0 0f 9f 9f ab ae 2a 05 41 f0 de c0 a7>.....N.....(.....d.....>.....>.....A.....>.....
03f0 f2 5e e0 74 ab ad 2d c0 c2 ad>.....>.....t.....N.....b.....>.....

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